ธนาคารแห่งประเทศไทย
BANK OF THAILAND



Monetary Policy Report

June

2020

Monetary Policy Report

The *Monetary Policy Report* is prepared quarterly by staff of the Bank of Thailand with the approval of the Monetary Policy Committee (MPC). It serves two purposes: (1) to communicate to the public the MPC's consideration and rationales for the conduct of monetary policy, and (2) to present the latest set of economic and inflation forecasts, based on which the monetary policy decisions were made.

The Monetary Policy Committee
June 2020

Mr. Veerathai Santiprabhob	Chairman
Mr. Mathee Supapongse	Vice Chairman
Mr. Paiboon Kittisrikangwan	Member
Mr. Sethaput Suthiwart-Narueput	Member
Mr. Kanit Sangsubhan	Member
Mr. Subhak Siwaraksa	Member
Mr. Somchai Jitsuchon	Member

Monetary Policy in Thailand

Monetary Policy Committee

Under the Bank of Thailand Act, the Monetary Policy Committee (MPC) comprises the governor and two deputy governors, as well as four distinguished external members representing various sectors of the economy, with the aim of ensuring that monetary policy decisions are effective and transparent.

Monetary Policy Objective

The MPC implements monetary policy under the flexible inflation targeting regime. While regarding medium–term price stability as its primary objective, the MPC also aims at supporting sustainable, full-potential economic growth and preserving financial stability, attributing to long-term price stability and economic sustainability

Monetary Policy Target

On December 24, 2019, the Cabinet approved the monetary policy target for 2020, which was mutually agreed between the MPC and the Minister of Finance to set the headline inflation within the range of 1–3 percent as the target for the medium-term horizon and for 2020.

In the event that average headline inflation in the past 12 months or a forecast of average headline inflation over 12 months ahead breaches the target range, the MPC shall send an open letter to the Minister of Finance to explain reasons for the breach of the target range, together with measures taken and estimated time to bring inflation back to the target. In addition, the MPC will write an additional open letter to the Minister of Finance every six months if average headline inflation based on the above criteria remains outside the target range.

Monetary Policy Instrument

The MPC utilizes the 1-day bilateral repurchase transaction rate as the policy interest rate to signal the monetary policy stance.

Evaluation of Economic Conditions and Forecasts

The Bank of Thailand takes into account information from all sources, the macroeconomic model, data from each economic sector, as well as surveys of large enterprises, together with small and medium-sized enterprises from all over the country, and various financial institutions to ensure that economic evaluations and forecasts are accurate and cover all aspects, both at the macro and micro levels.

Monetary Policy Communication

Recognizing the importance of monetary policy communication to the public, the MPC employs various channels of communication, both in Thai and English, such as (1) organizing a press statement at 14:00 on the day of the Committee meeting, (2) publishing edited minutes of the MPC meeting two weeks after the meeting, and (3) publishing the *Monetary Policy Report* every quarter

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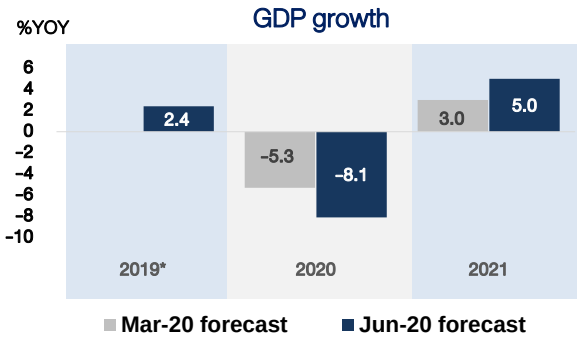
Economic assessment

Financial stability assessment

Note: Data in this report is as of June 23, 2020 (one day before the Monetary Policy Committee meeting)

Key Highlights of the Monetary Policy Report, June 2020

Economic growth projection



Note: * Outturn

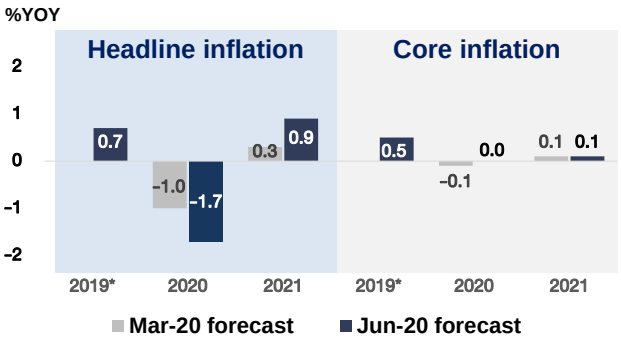
Source: NESDC, estimated by Bank of Thailand

Containment measures to control the spread of COVID-19 caused disruptions to global economic activities.

The Thai economy would gradually recover from second half of 2020 in tandem with many countries that could effectively contain the outbreak.

Public sector introduced relief measures that helped alleviate adverse impacts and would support the economic recovery going forward.

Inflation projection



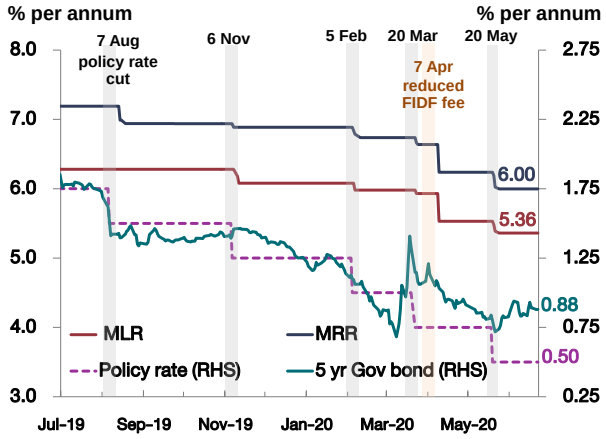
Note: *Outturn

Source: Ministry of Commerce, estimated by Bank of Thailand

Energy prices decreased due to a sharp decline in global crude oil prices.

Limited demand-pull was due to temporary disruptions of economic activities, high levels of excess production capacity.

Financial conditions



Source: ThaiBMA and Bank of Thailand (data as of 23 June 2020)

Note: MLR and MRR are averages of 5 largest commercial banks' lending rates

- The BOT's policy rate cuts and the reduction in the FIDF contribution resulted in continued decline in lending rates.
- Measures to stabilize financial markets (Mar 22, 2020) helped lower volatilities in financial market.

Financial stability

- Stability of financial institutions, external, and fiscal sectors, remained robust

- Sharp corrections in asset prices in financial markets
- Defaults by businesses and households in many countries including Thailand
- Corporate bonds being downgraded to non-investment grade

- Volatility in the Thai financial market decreased, but it should be closely monitored.
- Management of unsold inventories in real estate sector.

Executive Summary

Monetary Policy Conduct in the Second Quarter of 2020

At the meeting on May 20, 2020, the Monetary Policy Committee voted 4 to 3 to cut the policy rate by 0.25 percentage point from 0.75 to 0.50 percent. Three members voted to maintain the policy rate at 0.75 percent at this meeting. The Committee viewed that the Thai economy would experience a larger contraction in 2020 than previously assessed due to a more-severe-than-expected impact of the COVID-19 outbreak and the containment measures implemented by governments in many countries including Thailand which led to disruptions of global economic activities. Moreover, the impacts of the economic contraction on employment and financial system vulnerability would be more pronounced. For this reason, most members viewed that additional monetary policy accommodation was necessary to provide an effective and timely support to the real economy. However, three members deemed it appropriate to assess the effectiveness of the previously implemented financial and credit measures. **On June 24, 2020, the Committee voted unanimously to maintain the policy rate at 0.50 percent.** At this meeting, the Committee assessed that the Thai economy would contract in 2020 more than the previous assessment, as the impact of the COVID-19 outbreak turned out more severe than expected. However, a gradual recovery was expected in the second half of 2020, when activities should begin to rebound in countries that could effectively contain the outbreak, including Thailand. The Committee viewed that extra accommodative monetary policy since the beginning of the year as well as the fiscal, financial, and credit measures additionally announced helped alleviate adverse impacts and would support the economic recovery after the pandemic subsided.

Looking ahead, as the environment facing the Thai economy would remain highly uncertain, the Committee would thus stand ready to use additional appropriate monetary policy tools if necessary. Moreover, at the current juncture, the Committee deemed it important to (1) efficiently distribute liquidity in the financial system, which remained ample at the macroeconomic level, to businesses and households affected by the pandemic, (2) accelerate debt restructuring for borrowers while debt holidays and debt reduction measures were still in place, and (3) introduce supply-side policies to support changing economic structure in order to ensure a strong recovery in the post-COVID environment. These included, such as, establishing mechanism to manage excess production capacity, enhancing high-potential industries, accelerating infrastructure investments, facilitating upskill of labor, and accelerating reform of government regulations that remained obstacles.

Assessment of the Economic and Financial Outlook as the Basis for Policy Formulation

1. Global Economy

The global economy would severely contract as governments in many countries implemented containment measures, causing disruptions to global economic activities. Consequently, this severely affected manufacturing, employment, private consumption, and private investment, as well as merchandise exports, which contracted across major advanced economies and Thailand's trading partners in Asia. Thus, most countries especially major advanced economies were expected to enter a recession in 2020. **Given such development, the Committee revised down economic growth projections for Thailand's trading partners to a 4.6 percent contraction in 2020, followed by a 4.2 percent expansion in 2021. Also, there remained risks that trading**

partner economies would underperform the baseline projection. These risks included a more severe and prolonged outbreak than previously assessed due to an emergence of the second wave of the outbreak, greater vulnerabilities in the global financial system owing to increased risks of defaults or credit rating downgrades among corporates or governments in some countries, re-intensifying trade tensions between the U.S. and China, and geopolitical risks such as tensions between Mainland China and Hong Kong. On the other hand, there were also possibilities that trading partners' economic growth would outperform the baseline projection if the pandemic ended sooner than expected or government stimulus measures worldwide would turn out to be more effective than expected.

Governments in various countries worldwide implemented additional fiscal and monetary measures to alleviate the economic impact of the COVID-19 outbreak. Governments introduced series of relief measures aimed particularly at SMEs, households, and employment. At the same time, central banks in major advanced economies implemented unconventional measures in addition to maintaining ultra-low policy interest rates, while many regional central banks eased monetary policy further. Also, a number of central banks introduced measures to support stability of the financial markets to ensure sufficient liquidity in their respective markets.

2. Financial Conditions and Financial Stability

Overall financial conditions in Thailand became more accommodative as reflected by the real policy interest rate, computed using the one-year ahead inflation expectations, that declined and remained low compared with those in other emerging markets. In addition, interest rates in Thailand's financial markets declined. Short-term government bond yields fell by more than long-term yields, causing the government bond yield curve to steepen. Corporate credit spreads stabilized at a high level after having risen at the beginning of the second quarter of 2020, as investors were concerned about the profitability outlook of some businesses. Meanwhile, commercial bank lending rates declined substantially in line with the policy rate and the reduction in the Financial Institutions Development Fund (FIDF) contribution. Corporate financing accelerated in line with the need for liquidity buffer among businesses, and also as a result of the government's soft loan programs. With regard to exchange rates, the baht against the U.S. dollar and the effective exchange rate appreciated throughout the second quarter, thanks to Thailand's success in containing the spread of the virus. Adding to the strength of the baht were sales of the U.S. dollars by domestic players, particularly those related to the sales of gold when pandemic uncertainties drove up the gold price. Nevertheless, the Committee expressed concerns over the strength of the baht and its potential impacts on economic recovery. The Committee would thus closely monitor developments in the foreign exchange markets in order to assess the necessity of implementing additional appropriate measures.

The Thai financial system became more vulnerable due to this year's contraction of the economic outlook in light of the COVID-19 situation, although stability of financial institutions, the external sector, and the fiscal sector remained robust. A larger-than-expected contraction of the global economy could result in (1) sharp corrections in asset prices in global financial markets, (2) defaults by businesses and households in many countries including Thailand, and (3) corporate bonds being downgraded to non-investment grade.

3. Economic and Inflation Outlook

The Thai economy would contract by 8.1 percent in 2020, more than previously assessed in the previous *Monetary Policy Report*. Such severe contraction would be observed across almost all economic growth components, except public expenditure. This was due to a more-severe-than-expected COVID-19 pandemic and the containment measures implemented by governments in many countries including Thailand. This led to temporary disruptions of global economic activities especially in the second quarter of 2020. Thailand's tourism and merchandise exports were thus projected to markedly contract, which would weigh on private consumption as a result of weakening income and employment. A fall in domestic and external demand would also lead to a sharp contraction in private investment. However, public expenditure would play an important role in supporting the economic recovery in the period ahead. The support would come from the reallocation of budget structure for a faster disbursement to remedy and restore the economy, as well as an additional liquidity support from the third phase of government relief measures to alleviate the impact of COVID-19. **Nevertheless, the Thai economy was expected to bottom out in the second quarter of this year and would gradually recover from the third quarter onward,** following the relaxation of containment measures in many countries including Thailand. Merchandise exports, tourism, and domestic demand would therefore exhibit a smaller contraction. **In 2021, the Thai economy was projected to expand at 5.0 percent, higher than previously assessed.** This was partly due to the lower base in 2020, and as government was expected to launch additional employment stimulus package and large-scale investment projects. However, economic activities in Thailand would remain substantially below its potential level.

Additional assumptions underlying the economic projection included, first, there would be no severe second-wave of the COVID-19 outbreak in Thailand. This would allow for the gradual re-opening of the country under the travel bubble scheme, which would be done through partnerships between COVID-19-free countries, permitting travelling into targeted locations in each countries starting from the second half of 2020. Second, there would be a widespread availability of COVID-19 vaccines from the fourth quarter of 2021 onward.

The value of Thai merchandise exports in 2020 would experience a sharper decline than previously assessed, due mainly to contraction in volume. This was in line with the outlook of trading partner economies and global trade volume, as well as supply chain disruptions in some businesses. Meanwhile, prices of merchandise exports were projected to decline, mainly owing to prices of goods that were sensitive to movements in crude oil prices. Nevertheless, the gradual relaxation of containment measures in many countries would support the gradual recovery of merchandise exports from the third quarter of 2020 onward. In 2021, the value of merchandise exports was expected to expand, both in terms of volume in response to the recovery of global economy and global trade, and in terms of price in line with the outlook of global crude oil prices. However, the value of merchandise exports would be below the pre-pandemic level, as external demand had yet to fully recover. In addition, changing patterns of business operations worldwide under the new way of life in the post-COVID environment would limit the potential of certain industries to fully recover.

The tourism sector was projected to recover more slowly than in the previous assessment, as the outbreak in other countries would be more prolonged and widespread than previously expected. The Thai government was expected to maintain its travel restrictions on inbound foreign

tourists for an extended period and would begin to gradually re-open the country for foreign tourists from certain countries under the travel bubble scheme in the second half of 2020.

The current account balance was projected to register a surplus of 15.5 and 20.2 billion U.S. dollars in 2020 and 2021, respectively. The current account balance in 2020 was revised down due to a larger deficit in the services balance, mainly as a result of a marked decline in tourism receipts. For 2021, the current account balance was projected to increase from the previous year, owing to gradual improvements in trade balance and tourism receipts.

Public expenditure would play an important role in shoring up the economy in 2020 and supporting the recovery in the period ahead. The government was expected to accelerate disbursement of the fiscal year 2020 budget during the rest of the year, as well as reallocate the budget structure for a faster disbursement to remedy and restore the economy thanks to the Budget Expenditure Transfer Bill, B.E. 2563 (A.D. 2020) and the revision of the fiscal year 2021 budget structure. There would also be an additional liquidity support from the third phase of government relief measures to alleviate the impact of COVID-19. In addition, the government would introduce further measures to help those affected by the outbreak, which would partly help sustain business operations and employment. Nevertheless, investment spending by state-owned enterprises was projected to expand, which would be key mechanism to support the economy going forward.

Private consumption was expected to contract in 2020 more than the previous assessment, particularly in the second quarter. This was due to the containment measures, as well as a marked decline in non-farm income and weakening consumer confidence. However, the gradual relaxation of containment measures would lead to a gradual improvement in economic activities in the period ahead. Nevertheless, private consumption was expected to grow at a level below the pre-pandemic level throughout the forecast horizon. This was because households were expected to be cautious in spending given high uncertainties surrounding the COVID-19 situation going forward. Furthermore, households' financial positions would be more vulnerable given falling income and elevated debt levels.

Private investment was expected to contract in 2020 more than the previous assessment. This was on account of contractions in both domestic and external demand, as well as high levels of excess production capacity, which caused most businesses to postpone their investment plans. However, the government's infrastructure investment projects were underway, particularly the public-private partnership (PPP) projects both within and outside the Eastern Economic Corridor (EEC). As for 2021, some investment projects would likely continue, particularly large-scale infrastructure investment and infrastructure investment to support the digital economy.

Going forward, the Thai economy would continue to face high uncertainties. Risks to the economic projection tilted downward and depended largely on the impact of COVID-19. The possibilities that the Thai economy would underperform the baseline projection would be due to the following factors. First, the global economy could recover more slowly than expected or enter a severe recession, which would lead to a larger-than-expected impact on Thai merchandise exports. Second, default risks of businesses and households could increase. Third, the recovery of the tourism sector could be slower than expected. Forth, Thailand could experience the second wave of the COVID-19 outbreak, triggering another round of lockdown measures. However, **there were possibilities that the Thai economy**

would outperform the baseline projection due to (1) better-than-expected impacts of additional government's stimulus measures to shore up domestic consumption and investment, and measures to assist borrowers, (2) public infrastructure investment and PPP projects which could incentivize the private sector to invest more than expected, and (3) more relocations of production base to Thailand thanks to the success in containing the spread of the virus.

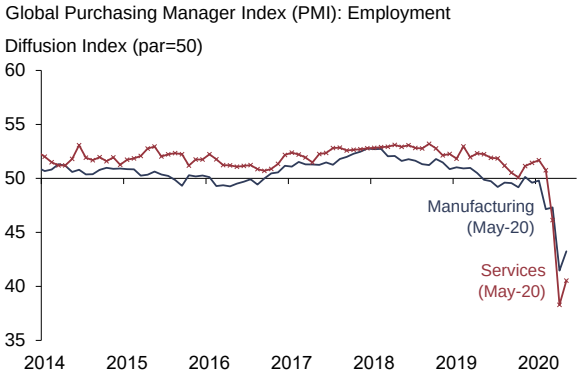
Headline inflation in 2020 would be more negative than previously assessed, mainly due to supply-side factors. This was particularly because of sharply declined energy prices in line with global crude oil prices, which was owing to decreasing demand following disruptions of global economic activities and high levels of excess production capacity. Moreover, a fall in headline inflation was partly due to government measures to reduce electrical and water bills. Meanwhile, **core inflation would remain low** in line with muted demand-pull inflationary pressures. **Looking ahead, headline inflation would gradually rise toward the target** mainly in tandem with gradually rising crude oil prices. However, core inflation would remain subdued in line with Thailand's gradual economic recovery outlook. **The Committee thus projected headline inflation to be negative at 1.7 percent in 2020 and turn positive at 0.9 percent in 2021. Core inflation was projected to be zero and 0.1 percent in 2020 and 2021, respectively.** Going forward, the inflation outlook would remain highly uncertain, as the COVID-19 pandemic had altered global economic structure, which resulted in large excess capacity in several industries. In addition, trade protectionism and geopolitical risks could lead to deglobalization, which would eventually result in a significant impact on global supply chains and prices of goods.

1. Global Economy

Governments in many countries implemented containment measures to control the spread of COVID-19, causing disruptions to global economic activities. This severely affected economic prospects of Thailand's trading partner economies, particularly major advanced economies which were expected to enter a recession this year. Nevertheless, a series of financial and fiscal measures that had been implemented would help alleviate adverse impacts and support the economy in the period ahead.

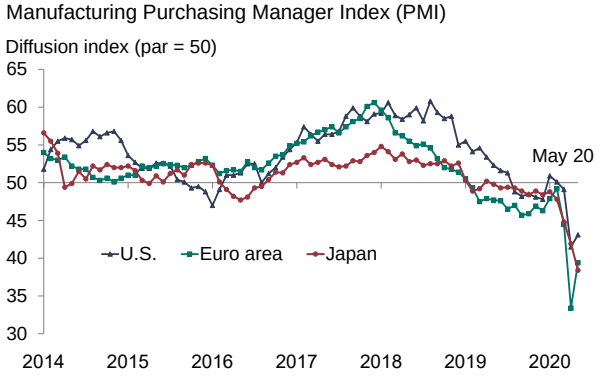
In the first quarter of 2020, major advanced economies contracted as a result of containment measures in the U.S., the euro area, and Japan. Consequently, major advanced economies contracted from the previous quarter due to manufacturing, private consumption, and private investment, as well as merchandise exports, which contracted in line with global demand.

Chart 1.1 Global employment in the manufacturing and services sector was expected to sharply decline



Source: JP Morgan

Chart 1.2 G3's manufacturing sector was expected to contract in the period ahead



Sources: Bloomberg and Eurostat

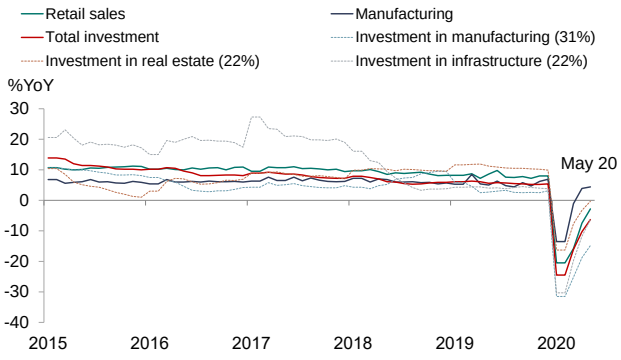
In 2020, major advanced economies were projected to continue to contract substantially and enter a recession. In particular, the economies would exhibit a sharper contraction in the second quarter of 2020 due to the severe outbreak situation, prompting governments to continue implementing containment measures to control the spread of the virus. However, the economies would gradually recover in the second half of 2020 following the gradual relaxation of containment measures. Meanwhile, private consumption was expected to recover slowly as consumers became more cautious in their spending due to several factors. These included, first, weak labor markets (Chart 1.1), and, second, changes in spending behavior under the new normal, such as work from home, resulting in less commutes and away-from-home spending. Moreover, tourism-related businesses would not recover fully in the initial phase, since tourism under the new normal following the relaxation of travel restrictions would take the form of a travel bubble, which would be done through partnerships between COVID-19-free countries, permitting travelling into targeted locations in each country. Third, weak consumer confidence due to an unfavorable income prospect and a protest in the U.S. that raised concerns over the second wave of the outbreak could delay the relaxation of containment measures. Meanwhile, production was expected to increase slightly (Chart 1.2) after businesses were able to resume their operations. However, capacity utilization would remain lower than the pre-pandemic level, as global demand remained weak and the recovery of global trade was still weighed down by the U.S.-China

trade tensions, which could intensify again. Nevertheless, a series of financial and fiscal measures previously announced would help mitigate impacts on the economy and reduce risk of a severe contraction in the period ahead.

The Chinese economy was projected to expand at a slower pace, while Asian economies would contract as the COVID-19 containment measures resulted in disruptions to economic activities across several sectors. However, a series of financial and fiscal measures implemented would help alleviate the impacts and shore up the economy in the period ahead.

In the first quarter of 2020, the Chinese and Asian economies (excluding Japan) contracted from the previous quarter, since the COVID-19 outbreak and the containment measures caused manufacturing, private consumption, private investment, and exports to contract. **In 2020, the Chinese economy would exhibit a more modest growth than previously assessed** due to strict compliance to social distancing measures, although businesses gradually resumed their operations after the Chinese authorities announced a relaxation of containment measures since the beginning of the second quarter (Chart 1.3). In addition, external demand contracted more than expected as a result of containment measures in major trading partner countries, such as the U.S., the euro area and Japan. Going forward, falling employment and income would weigh on the recovery of the Chinese economy. However, a series of government measures that had been implemented would help mitigate the impact of the outbreak, preserve financial stability, as well as reduce risk of recession for the Chinese economy.

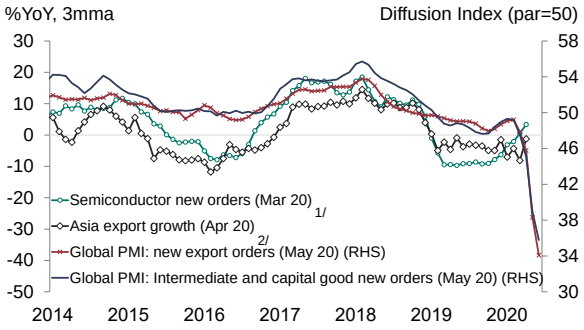
Chart 1.3 The Chinese economy would expand at a slower rate due to a slower-than-expected economic recovery, although businesses gradually resumed their operations after the relaxation of containment measures



Note: ( ) denotes share to total investment
Source: CEIC

Chart 1.4 Global trades were expected to recover more slowly than expected, as reflected by new orders for capital, intermediate and export goods

Asian leading export indicators



Note: 1/ Semiconductor new orders are calculated from World semiconductor billing and Taiwan's tech export orders
2/ Asian export growth is calculated from export growth of China, Hong Kong, Taiwan, South Korea, Malaysia, Singapore, Indonesia and the Philippines
Sources: CEIC, JP Morgan

In 2020, Asian economies (excluding China and Japan) were projected to contract, despite a relaxation of containment measures in some countries. This was because many countries were still coping with a growing number of COVID-19 cases, while some countries faced the second wave of the outbreak after relaxing the containment measures, such as Singapore, Indonesia, the Philippines, and South Korea. Moreover, strict social distancing measures in many countries continued to weigh on private consumption, services sector, and tourism. Meanwhile, the manufacturing and export sectors would recover slowly, but signs of new orders for electronics products were observed partly due to temporary

pent-up demand driven by work-from-home. Going forward, the electronics cycle would recover at a somewhat slower pace than previously assessed (Chart 1.4), mainly due to weakening global demand and investment. However, a series of stimulus measures in many countries would help reduce the risk of Asian economies experiencing a sharp contraction in the period ahead.

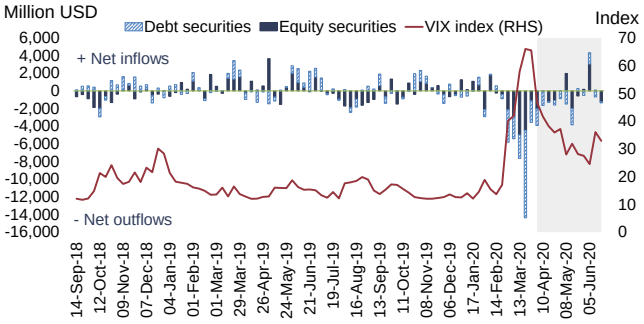
Net capital outflows declined, with inflows resuming in both equity and bond markets of emerging markets in some periods, as concerns among investors in the global financial markets subsided.

In the second quarter of 2020, concerns among investors in the global financial markets subsided. This was reflected in a continued decline in the VIX index[1/] (Chart 1.5) since the end of March 2020. This was on account of accommodative monetary policies, particularly those implemented by advanced economies' central banks, additional fiscal measures that had been introduced in many countries to remedy and restore the economy, the relaxation of containment measures, as well as the progress on vaccines development. These factors helped restore investor confidence and led to a more gradual sell-off in both equities and bonds of emerging markets. At the beginning of June 2020, investors had resumed their investment in risky assets, particularly equity markets in emerging Asia, as reflected in rising MSCI EM index (Chart 1.6). However, investor confidence would likely remain volatile due to development of the COVID-19 pandemic situation, as well as the economic data outturns in each period.

Looking ahead, capital flows to emerging markets would continue to be highly volatile, due to the prolonged COVID-19 outbreak in many countries and risks of the second wave of the outbreak, re-intensifying trade tensions between the U.S. and China, geopolitical risks such as tensions between Mainland China and Hong Kong, as well as risk to global financial system stability owing to rising solvency risks among corporates and households.

Chart 1.5 Net capital outflows declined, with inflows resuming in both bond and equity markets of emerging markets in some periods, as concerns among investors in the global financial markets subsided

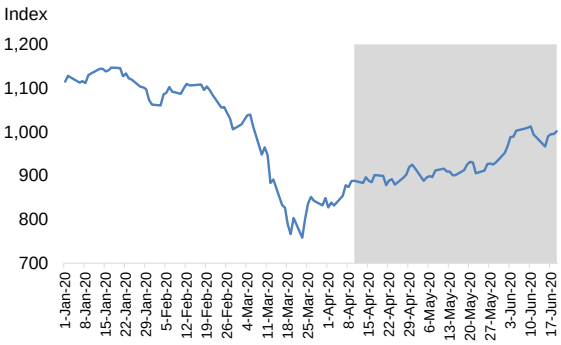
Net capital inflows to EMs* and VIX index (weekly)



Note: *EMs include Thailand, Indonesia, India, South Africa, and Turkey
Sources: Bloomberg and Institutional Institute of Finance

Chart 1.6 Investors resumed their investments in risky assets, particularly in equity markets in emerging Asia since the beginning of June 2020

Morgan Stanley Capital International Emerging Markets Index (MSCI EM)



Source: Bloomberg

[1/] VIX Index is a measure of the stock market volatility implied by S&P 500 Index options.

Governments in many countries worldwide implemented additional financial and fiscal measures to alleviate the economic impact of the COVID-19.

Governments in many countries worldwide implemented additional financial and fiscal measures to alleviate the economic impact of the COVID-19. Governments in major advanced and emerging market economies introduced a series of measures to shore up the economy, including relief measures aimed particularly at SMEs, households, and employment[2/]. However, governments in many major advanced economies started to face limitations in introducing additional measures due to lower fiscal policy space, as reflected in historically high public-debt-to-GDP ratios. Meanwhile, **central banks in major advanced economies implemented unconventional measures in addition to maintaining ultra-low policy interest rates.** The Federal Reserve (Fed) continued its quantitative easing (QE) measures, while the European Central Bank (ECB) continued to inject liquidity in the financial system through longer-term refinancing operations (LTROs), and also undertook targeted longer-term refinancing operations (TLTROs) in order to support credit extension by financial institutions to the real economy. At an unscheduled meeting in May 2020, the Bank of Japan (BOJ) launched a New Fund-Provisioning Measure to provide liquidity support for businesses, particularly SMEs. Meanwhile, **many regional central banks eased monetary policy further** by lowering their policy rates in order to mitigate the impact of the outbreak on the economy. These regional central banks included, for instance, the Bangko Sentral ng Pilipinas (BSP), Bank Negara Malaysia (BNM), and Bank of Korea (BOK). At the same time, some central banks extended the period of measures to alleviate financial burdens for borrowers, such as temporary debt holidays and soft loans offering low interest rates for financial institutions to encourage lending to the private sector.

Moreover, a number of central banks introduced measures to safeguard stability in the financial markets to ensure sufficient liquidity in their respective markets. At the June-2020 meeting, the Fed stated that it would inject liquidity in the bond market through additional purchases of government bonds, agency mortgage-backed securities, as well as purchases of corporate bonds via exchange traded funds (ETFs) and direct purchases, to ensure the well-functioning of the financial markets and enhance the effectiveness of monetary policy transmission. Meanwhile, the ECB announced additional purchases of government and corporate bonds through the Pandemic Emergency Purchase Program (PEPP), with the increased total amount from 750 to 1,350 billion euros. Regional central banks also announced additional measures to ensure stability in the financial markets. These included, for instance, increased volume of government bond purchases, and injecting liquidity into the financial system via other instruments such as lowering reserve requirement ratio (RRR).

Thailand's trading partner economies were projected to contract more severely in 2020 as a result of a-more-severe-than-expected impact of the COVID-19 containment measures, particularly in the second quarter of 2020. However, the economies were expected to resume growth in 2021. Nonetheless, risks that trading partner economies would underperform the baseline projection increased mainly because the COVID-19 pandemic could become more severe and prolonged than expected.

[2/] For more information, please see the link COVID-19 and Government Measures in Foreign Countries, May 2020 (Available in Thai only).

Thailand's trading partner economies were projected to contract in 2020 more than previously assessed in the previous *Monetary Policy Report*, as the pandemic would be prolonged and weigh on economic activities more than expected. Moreover, containment measures in many countries severely affected consumption, investment, manufacturing, and exports. Moreover, the services sector was substantially affected and was expected to recover more slowly than the manufacturing sector. However, financial and fiscal measures that were already implemented and forthcoming would partly help prevent a severe economic contraction. In 2021, trading partner economies were expected to recover, as the COVID-19 outbreak would gradually subside, resulting in a gradual relaxation of containment measures. **The Committee thus revised down economic growth projections for Thailand's trading partners to a 4.6 percent contraction in 2020, followed by a 4.2 percent expansion in 2021.** (Table 1.1).

Table 1.1 Assumption on trading partner growth

Annual change (%YoY)	Weight (%)	2019*	2020	2021
United States	18.2	2.3	-7.0 (-0.3)	3.2 (-1.4)
Euro area	9.9	1.2	-10.0 (-2.6)	4.2 (-0.3)
Japan	14.3	0.7	-5.4 (-1.7)	1.7 (0.7)
China	17.0	6.1	0.9 (3.9)	8.9 (5.5)
Asia (excluding Japan and China)*	32.4	2.7	-3.9 (0.2)	3.6 (2.1)
Total***	100	2.7	-4.6 (0.2)	4.3 (2.0)

Note: *Outturn

**Weighted by 7 trading partners' shares in Thailand's exports in 2019, namely, Singapore (5.1%), Hong Kong (6.8%), Malaysia (6.1%), Taiwan (2.3%), Indonesia (5.3%), South Korea (2.7%), and Philippines (4.0%)

***Weighted by 13 trading partners' shares in Thailand's exports in 2019 (including the United Kingdom and Australia)

( ) as reported in Monetary Policy Report, March 2020

The Committee assessed that there remained risks that trading partner economies would underperform the baseline projection. These risks included (1) a more severe and prolonged COVID-19 outbreak than previously assessed due to an emergence of the second wave of the outbreak, which could lead many countries to re-introduce the containment measures, (2) greater vulnerabilities in the global financial system owing to increased default risks among corporate and governments in some countries, or credit-rating downgrades, (3) re-intensifying trade tensions particularly between the U.S. and China, and (4) geopolitical risks such as tensions between Mainland China and Hong Kong. **On the other hand, there were also possibilities that trading partners' economic growth would outperform the baseline projection** owing to (1) a faster-than-expected ending of the outbreak due to effective healthcare measures in many countries and successful development of vaccines and medicines for the COVID-19, and (2) larger-than-expected effects of financial and fiscal measures in many countries in stimulating the economy.

Dubai crude oil prices in the second quarter of 2020 continued to decline on the back of weakening global demand. However, crude oil prices rose slightly toward the end of the second quarter following an agreement on a crude oil production cut among major oil producers.

Dubai crude oil prices in the second quarter of 2020 continued to decline from the previous quarter. During late March and May, crude oil prices dropped below 30 U.S. dollars per barrel, due to a sharp fall in global demand as a result of the COVID-19 outbreak and a significant increase in crude oil production. This was because Saudi Arabia boosted its crude oil production after Russia refused to agree to the OPEC's proposal at the meeting in March, thereby triggering market concerns over global crude oil storage capacity. **However, crude oil prices gradually increased toward the end of the second quarter, as OPEC and its allies such as Russia and the U.S. reached an agreement on a crude oil production cut in April 2020.** Moreover, in June 2020, Saudi Arabia, Kuwait, and the United Arab Emirates pledged additional production cuts according to the agreement, while the U.S. reduced their production further due to breakeven cost considerations. As a result, global crude oil production fell more than expected. Going forward, however, crude oil prices would continue to be weighed down by negative factors, such as elevated crude oil inventory level and subdued global oil demand as a result of the COVID-19 pandemic, particularly for fuels used in transportation and travel.

Despite supply-side factors causing larger-than-expected production cuts, crude oil prices would continue to be weighed down by crude oil inventory level and global demand. The Committee therefore maintained the projection of Dubai crude oil prices in 2020 at 35.1 U.S. dollars per barrel, close to the previous projection at 35.0 U.S. dollars per barrel. In 2021, Dubai crude oil prices were projected to increase from 34.0 to 37.0 U.S. dollars per barrel (Chart 1.7), where crude oil prices would gradually increase in line with gradual recovery of the global economy and a reduction in crude oil supply from high-cost producers such as the U.S. Given high uncertainties surrounding the current situation, **risks to the projection of Dubai crude oil prices were expected to tilt downward**, as the COVID-19 outbreak could affect the global economy more than expected. Moreover, there were risks of re-intensifying trade tensions between the U.S. and China, and the possibility that OPEC and related countries would fail to meet the production cut agreement. On the other hand, Dubai oil prices could be higher than the assessment as a result of more-severe-than-expected geopolitical risks.

Chart 1.7 Dubai crude oil prices in Q2/2020 fell consecutively from the previous quarter mainly due to significant decline in demand as a result of the COVID-19 outbreak. However, the prices slightly increased toward the end of the second quarter due to the acceleration in production cut according to the agreement among major oil producers



2. The Thai Economy

2.1 Recent Developments

The COVID-19 outbreak became more severe and widespread, causing the Thai economy to contract in the first quarter of 2020 due to both external and domestic demand. The Thai economy was expected to contract substantially in the second quarter of 2020 due to strict containment measures imposed in Thailand and abroad.

In the first quarter of 2020, the Thai economy contracted 1.8 percent from the same period last year due to external and domestic demand slowdown in light of the COVID-19 outbreak that became more severe and widespread. Governments in many countries, including Thailand, thus implemented containment measures to control the spread of the virus, resulting in temporary disruptions of economic activities across several sectors. Exports of services fell sharply due to international travel restrictions, which led tourism spending and foreign tourist figures across all nationalities to decline substantially. Merchandise exports expanded on account of gold exports. However, merchandise exports (excluding gold) continued to decline due to the following factors. First, the COVID-19 outbreak dampened demand from trading partners and global trade volume. Second, international travel restrictions resulted in disruptions of supply chain and shipping routes. Third, the drought affected exports of agricultural products. Nonetheless, higher demand contributed to the export growth of certain products, such as food, medical products, and electronics, thanks to work-from-home measure and China's production capacity that had yet to fully recover. Domestic demand declined as household income and consumer confidence continued to weaken. Private consumption thus moderated despite growth in consumption of non-durable goods, especially necessities and electricity. Meanwhile, the delayed enactment of the Annual Budget Expenditure Act, B.E. 2563 (A.D. 2020) caused a continued contraction of public expenditure, both current and capital expenditures. However, investment of state-owned enterprises (SOEs) could still expand in line with investment in transportation and public utilities, while manufacturing and private investment declined on account of the contraction in economic activities.

In the second quarter of 2020, latest indicators pointed to a severe economic contraction from the previous quarter due to strict containment measures imposed worldwide. However, a series of government relief measures for those affected by the outbreak and the gradual relaxation of containment measures would shore up the economy in the period ahead. The more severe and global widespread COVID-19 outbreak led to temporary disruptions of economic activities in most trading partners. No foreign tourists were expected to visit Thailand at least until the end of June due to restrictions on inbound foreign tourists to control the spread of the virus. Merchandise exports (excluding gold) and manufacturing in almost all industries would contract markedly on account of a sharp decline in economic activities and global trade volume. Private consumption would contract in line with lower income of workers and concerns over the outbreak. Likewise, private investment would continue to fall due to considerably weaker demand and investor confidence, as well as high levels of excess production capacity. Nevertheless, public expenditure would increase on account of current expenditure due to disbursement for consumption of goods and services used in the public health response to prevent and control the outbreak. Public expenditure and

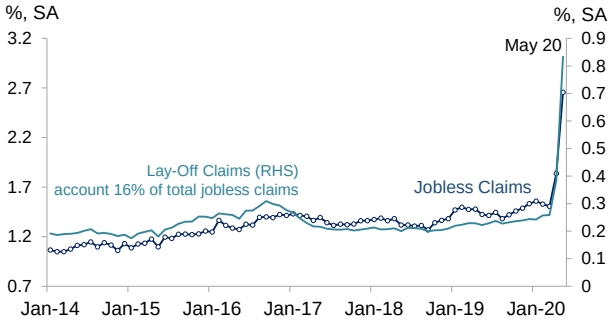
the gradual relaxation of containment measures would thus shore up the economy in the period ahead.

Overall household purchasing power significantly declined and became more vulnerable due to the COVID-19 outbreak and the drought.

The COVID-19 outbreak and containment measures caused household purchasing power to further decline and become more vulnerable. **Non-farm income** fell substantially as reflected in a marked decline in the number of insured workers in the social security system (Section 33 of the Social Security Act) across almost all sectors. This was accompanied by a significant increase in the number of workers who applied for unemployment benefits (Section 38) mainly due to the termination of employment (Chart 2.1). The number of employees and firms affected by the temporary suspension of business (Section 75) also increased substantially (Chart 2.2). This suggested a more severe loss in income and working hours of those employed in both manufacturing and services sector, particularly in hotel, tourism and trade businesses.

Chart 2.1 Non-farm income fell substantially as reflected in a significant increase in the number of workers who applied for unemployment benefits

Ratio of Jobless Claims to Total Contributors in Section 38*



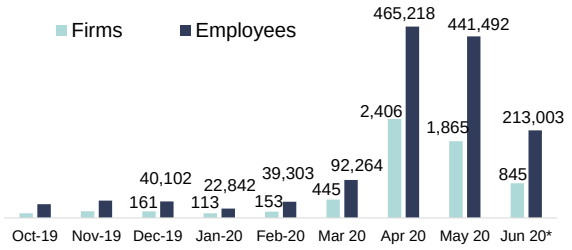
May 2020 : **Total Jobless Claims of 332,060 positions**, consist of lay-off claims of 105,488 postions, voluntary unemployment claims of 215,477 positions, and contract termination claims of 11,095 positions.

Note: excluding umployment caused by the COVID-19

Source: Social Security Office, calculated by Bank of Thailand

Chart 2.2 The number of employees and firms affected by the temporary suspension of business (Section 75) also increased substantially.

The number of firms and employees registered for the temporary suspension of business (Section 75)



Note : The Section 75 data is based on the total number of firms and employees reported to the Ministry of Labor each month.

*data as of June 7, 2020

Source: Department of Labor Protection and Welfare, Ministry of Labor

Farm income would fall on account of a significantly lower agricultural output, especially rice and sugarcane, mainly as a result of the drought. Agricultural prices would also fall, particularly prices of rubber due to weaker external demand. However, prices of white rice would increase due to lower output and higher external demand from the stocking up of food during the outbreak and the export quota imposed by major rice exporting countries. Nevertheless, there remained risk that warranted continued monitoring, including weather conditions given low water levels in dams and possible delay in rainfall. However, rainfall was expected to increase somewhat during the beginning of the rainy season.

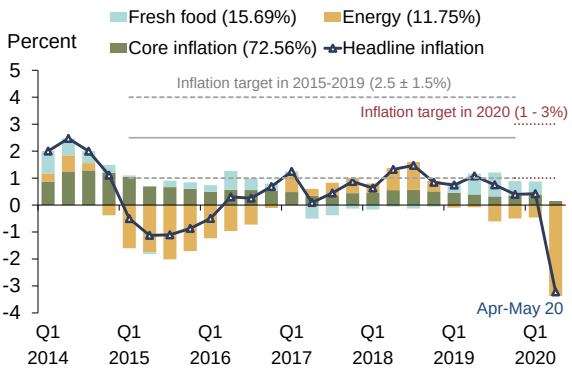
Headline inflation in April-May 2020 turned negative mainly due to energy prices, but was projected to be less negative in June due to gradually rising energy and fresh food prices.

Headline inflation in the first two months of the second quarter of 2020 averaged at -3.22 percent, down from 0.42 percent in the previous quarter (Chart 2.3). The fall in headline inflation was mainly in tandem with a larger decline in energy prices from the same period last year.

Domestic retail oil prices decreased in line with global crude oil prices and the government relief measures to cut electricity charges during the COVID-19 outbreak. Fresh food inflation also declined from the previous quarter. In particular, prices of vegetables and fruits fell significantly in line with domestic and external demand, as well as due to the higher base of vegetable prices in the previous year. In June, headline inflation was projected to be less negative due to gradually rising energy and fresh food prices, as global crude oil prices and demand were expected to gradually improve following the relaxation of containment measures.

Chart 2.3 Headline inflation decreased from the previous quarter mainly due to energy prices.

Headline inflation and inflation target

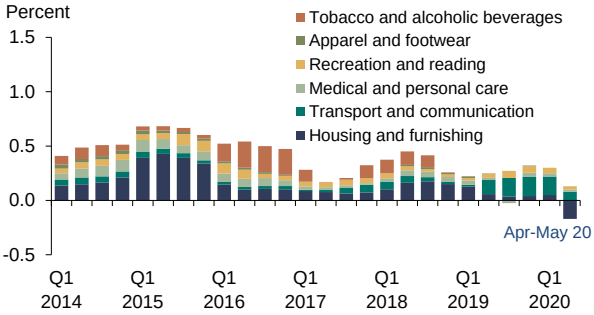


Note: Headline inflation for the first 11 months of 2019 recorded 0.69 percent
( ) denotes share in inflation baskets
Source: Ministry of Commerce, calculated by Bank of Thailand

Core inflation in the first two months of the second quarter of 2020 averaged at 0.21 percent, down from 0.53 percent in the previous quarter, mainly because of the fall in housing prices (Chart 2.4) in line with government relief measures to cut water bills. In addition, public transport prices grew at a lower rate due to the high base effect from last year's public transportation fare increase. Meanwhile, core inflation in the food category increased slightly owing to higher prices of seasoning and condiments (Chart 2.5).

Chart 2.4 Core inflation in the non-food category (72% of core inflation) slightly declined mainly due to water supply prices

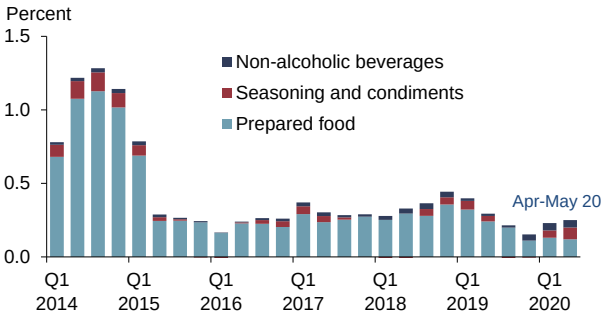
Contribution* to core inflation in the non-food category



Note: *Contributions to core inflation decompose core inflation into inflation of each component within the basket, weighted by its corresponding share.
Source: Ministry of Commerce, calculated by Bank of Thailand

Chart 2.5 Core inflation in the food category (28% of core inflation) increased on account of a rise in seasoning and condiment prices

Contribution* to core inflation in the food category



Note: *Contributions to core inflation decompose core inflation into inflation of each component within the basket, weighted by its corresponding share.
Source: Ministry of Commerce, calculations by Bank of Thailand

Table 2.1 Inflation

Annual percentage change	2018				2019				2020	
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Apr-May
Headline Consumer Price Index (Headline CPI)	0.64	1.31	1.47	0.84	0.74	1.08	0.61	0.40	0.42	-3.22
Core Consumer Price Index (Core CPI)	0.61	0.76	0.78	0.71	0.62	0.54	0.45	0.47	0.53	0.21
Fresh food	-1.04	-0.35	-0.82	-0.35	2.51	4.98	5.79	3.58	3.13	0.03
Energy	3.01	7.30	9.11	3.39	-0.78	-0.67	-4.95	-4.09	-3.98	-28.35

Source: Bureau of Trade and Economic Indices, Ministry of Commerce

Short-term (one-year ahead) inflation expectations edged up due to the very low base effect in the second quarter of 2020. According to the survey of professional forecasters in June 2020, inflation expectations recorded 1.3 percent, an increase from 1.1 percent in the

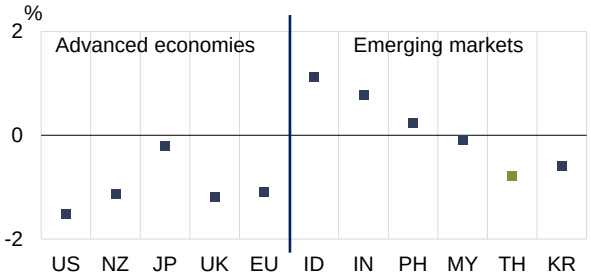
previous quarter. Inflation expectations based on the survey of businesses in May 2020 remained stable at 1.5 percent. **Long-term (five-year ahead) inflation expectations** according to the survey of professional forecasters in April 2020 declined from 2.0 percent in October 2019 to 1.8 percent. In addition, long-term inflation expectations based on a macro-finance term structure model with bond yields and macroeconomic data stabilized at 1.9 percent in the first quarter of 2020.

Short-term money market rates, lending rates, and deposit rates continued to fall in line with the policy rate, as well as the reduction in Financial Institutions Development Fund (FIDF) contribution.

During the second quarter of 2020, financial markets became more stable following a series of measures to stabilize financial markets introduced by the Bank of Thailand (BOT). These measures included the Mutual Fund Liquidity Facility (MFLF) and the establishment of Corporate Bond Stabilization Fund (BSF).[3/] At the same time, the Committee continued to cut the policy rate to the historically lowest level of 0.50 percent, which helped ease financial conditions further. The real policy rate fell from the previous quarter and remained low relative to those of other emerging market economies (Chart 2.6).

Chart 2.6 Thailand's real policy rate* decreased due to the policy rate cut and remained at a low level relative to those of other EM countries

Real policy rates*



Note: *Calculated from policy rate as of June 23, 2020 subtracted by one-year-ahead inflation expectation according to the survey by Consensus Forecasts (data as of June 8, 2020)
Sources: Bloomberg and Consensus Forecasts, calculated by Bank of Thailand

Short-term money market rates continued to decline in tandem with the policy rate cuts. **Short-term government bond yields** gradually fell since the beginning of the second quarter, primarily due to the policy rate cuts and increased demand for government bonds from mutual funds and contractual saving institutions (Chart 2.7). **Long-term government bond yields** declined by a lesser degree than short-term government bond yields, resulting in a steeper government bond yield curve. Long-term government bond yields moved in tandem with the U.S. treasury and regional bond yields, which were sensitive to economic figures and the COVID-19 outbreak situation at a given time.

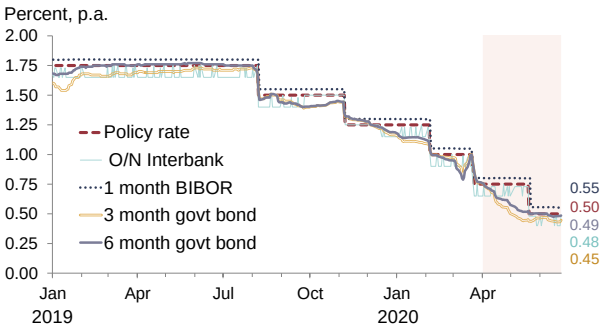
In the first week of the second quarter, government bond yields increased rapidly from the previous quarter as concerns over the COVID-19 outbreak prompted domestic mutual funds and foreign investors to sell off a considerable number of government bonds. However, government bond yields subsequently declined owing to concerns over the outlook of both Thai and foreign economies, before increasing again in late May following the relaxation of containment measures in many countries. Nonetheless, government bond yields became volatile toward the end of the quarter due to several factors, such as (1) investor confidence over the spread of COVID-19 and containment measures imposed in many countries, as well

[3/] The BSF was established under the Emergency Decree on the Maintenance of Stability of the Financial System and Economic Security of the Country, B.E. 2563 (2020). For more information, please visit: https://www.bot.or.th/English/FinancialMarkets/Pages/BSF.aspx

as progress on development of vaccines, and (2) the economic growth outlook of other countries (Chart 2.8).

Chart 2.7 Short-term money market rates continued to decline in tandem with one-time policy rate cut in Q2/2020

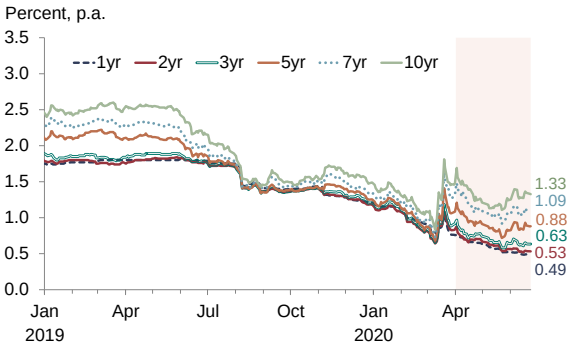
Short-term rates in financial market



Sources: Bank of Thailand and Thai Bond Market Association (Thai BMA) (data as of June 23, 2020)

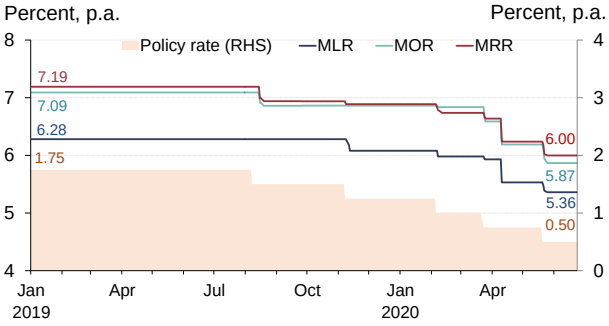
Chart 2.8 Short-term government bond yields gradually fell while long-term government bond yields declined by a lesser degree, resulting in a steeper government bond yield curve

Government bond yields



Source: Thai Bond Market Association (Thai BMA) (data as of June 23, 2020)

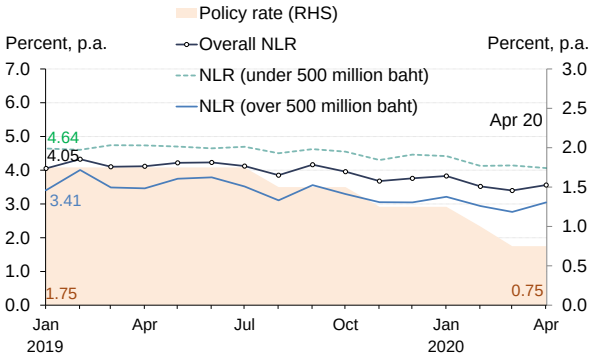
Chart 2.9 Commercial banks reduced their benchmark lending rates namely MLR, MOR and MRR in line with the policy rate and the reduction of FIDF contribution.



Note: Averages of 5 largest commercial banks' lending rates

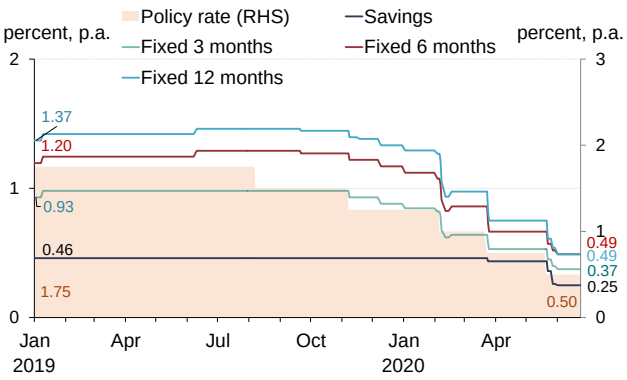
Source: Bank of Thailand (data as of June 23, 2020)

Chart 2.10 Overall new loan rates (NLR) of commercial banks remained low



Source: Bank of Thailand (data as of June 23, 2020)

Chart 2.11 Commercial banks gradually reduced their deposit rates.



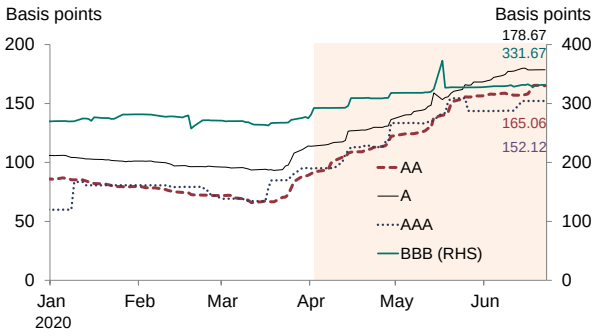
Note: Averages of 5 largest commercial banks' lending rates

Source: Bank of Thailand (data as of June 23, 2020)

The cost of financing through financial institutions fell in line with the policy rate and the reduction in the FIDF contribution from 0.46 to 0.23 percent of the deposit base for two years. The five largest commercial banks reduced lending rates quite significantly in this quarter. The average minimum loan rates (MLR) decreased to 5.36 percent, the average minimum retail rates (MRR) declined to 6.00 percent, and the average minimum overdraft rates (MOR) fell to 5.87 percent (Chart 2.9). Meanwhile, the new loan rates (NLR)[4/] in April 2020 remained low at 3.6 percent (Chart 2.10). Similarly, **deposit rates** gradually declined in tandem with reference loan rates (Chart 2.11). In particular, 3-month, 6-month and 12-month fixed deposits rates fell to 0.37, 0.49 and 0.49 percent respectively. Meanwhile, commercial banks reduced savings rate to 0.25 percent from 0.46 percent, level of which had been maintained since late 2015.

Despite the overall decline in government bond yields and commercial banks' rates, **corporate credit spread** remained stable at high levels following the increase in the beginning of the second quarter of 2020 (Chart 2.12). This was because large investors rapidly redeemed their investment units in mutual funds, especially those with majority of investment in foreign debt instruments, amidst excessively volatile foreign asset prices caused by the COVID-19 pandemic. Although the redemption slowed down following the BOT's introduction of the MFLF and volatility in the corporate bond market decreased following the establishment of the BSF, corporate bond yields remained high. This was because investors were still concerned over economic conditions as containment measures and sharply declined oil prices could weigh on the profitability of some businesses.

Chart 2.12 Corporate credit spread of 3-5 years remained stable at high levels.



Sources: Bank of Thailand and Thai Bond Market Association (Thai BMA) (data as of June 23, 2020)

Private sector financing accelerated on account of the need for liquidity buffer among businesses affected by the COVID-19 outbreak, and as a result of the government's soft loan programs for SMEs. Meanwhile, household credit growth continued to soften.

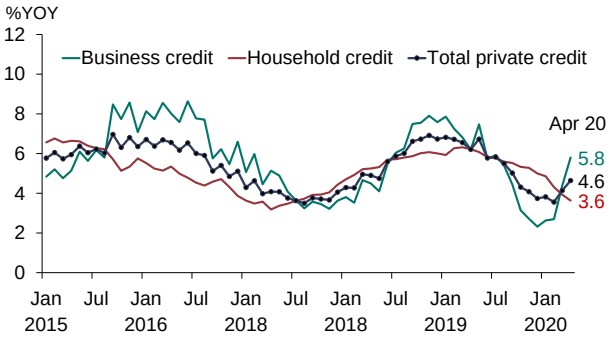
Since the first quarter of 2020, the BOT, related agencies, and financial institutions, had been introducing a series of measures to enhance liquidity and support the stability and well-functioning of financial markets. These measures helped ensure the distribution of liquidity in the financial system, which was ample at the macro level, to businesses and households affected by the pandemic, as well as relieve debt servicing burden in time of crisis. Thanks to these measures, the private sector financing could continue with details as follows.

[4/] NLR is the weighted average of interest rates on new loan contracts extended by 14 Thai commercial banks each month. The loan contracts exclude consumer loans, credit card loans, repurchase agreements, bank guarantees, as well as loans extended to financial intermediaries, the public sector and non-residents. The dataset covers loans with value of 20 million baht or higher for all loan types, purposes and maturities, and includes both secured and unsecured loans. Moreover, interest rates used in the calculation refer to the mid-rate between the lowest and the highest rates in each loan contract.

Private credits (private sector loans[5/] and bond issuances) continued to accelerate by 4.6 percent in April 2020 from to the end of the first quarter of 2019 (Chart 2.13). **Business credits** expanded 5.8 percent in April 2020 (Chart 2.14) on the back of a strong expansion in business loans. However, corporate bond issuances moderated, partly as investors became more cautious in investing in the corporate bond market after the period of liquidity strain at the end of March 2020. Thus, corporate financing through bond issuance was limited, especially those rated below BBB. In contrast, business loans accelerated in almost all sectors owing to (1) the need for extra liquidity to withstand the impact of the outbreak, as reflected in the surge in commercial credits especially for working capital (Chart 2.15), (2) the government's soft loan programs for SMEs, and (3) higher demand for loans to compensate for limited funding through bond issuances of some corporates. Meanwhile, **household credit growth slowed down further** to 3.6 percent, particularly credit card loans which softened on the back of a sharp fall in spending related to tourism, restaurants, and department stores.

Chart 2.13 Private credits (private sector loans and bond issuances) continued to accelerate while household credits continued to expand at a slower rate

Growth of private credit



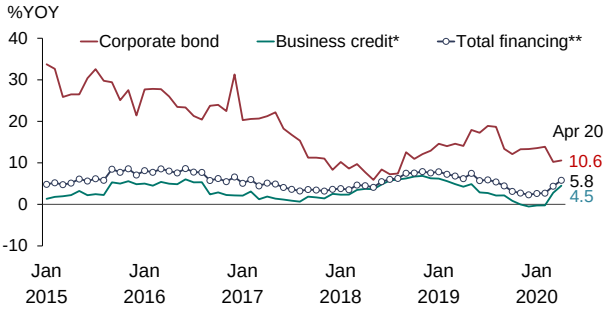
Note: 1) Private sector loans include credit extended by (1) Other Depository Corporations (ODCs) namely commercial banks, specialized financial institutions, finance companies, saving cooperatives, and money market mutual funds, (2) Other Financial Corporations (OFCs), and (3) non-residents

2) The definition of private credit was redefined to cover broader types of lenders and financial instruments. The data was first published in the MPR in 2019Q4, where the data was available from January 2012.

Source: Bank of Thailand

Chart 2.14 Overall business financing accerelated on account of business loans

Growth of business credit and corporate bond outstanding



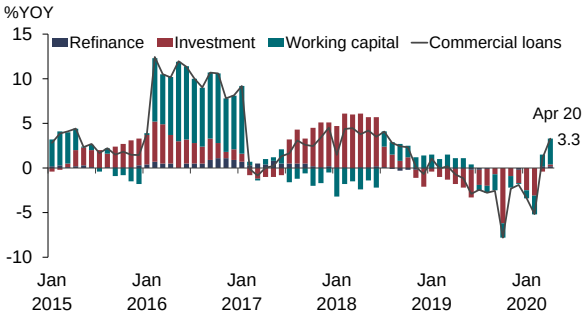
Note: * Business loans covers lending activities of Other Depository Corporation (ODCs), Other Financial Corporations (OFCs), and non-residents

** The definition of private credit was redefined to cover broader types of lenders and financial instruments. The data was first published in the MPR in 2019Q4, while the data was available from January 2012.

Sources: Thai Bond Market Association (Thai BMA) and Bank of Thailand

Chart 2.15 Business loan demand increased to be used as working capitals

Growth of commercial loan outstanding classified by loan purposes*



Note: *lending by Thai commercial banks (excluding their bank branches abroad) and foreign bank branches

Sources: Bank of Thailand

[5/] Loans extended to the private sector consist of those extended by (1) other depository corporations (ODCs) including commercial banks, specialized financial institutions, finance companies, saving cooperatives and market mutual funds, (2) other financial institutions (OFCs), and (3) non-residents.

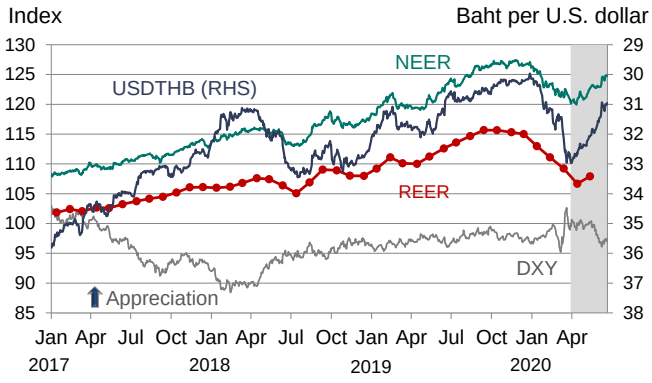
In the second quarter of 2020, demand for business credits was expected to continue rising as more liquidity would be needed to cushion against the economic impact of the COVID-19 pandemic, as well as to alter business models in the period ahead in response to people's new way of life. On the other hand, demand for household credits was expected to moderate since household income was affected by both the COVID-19 pandemic and the drought. Based on the survey in April 2020, financial institutions were expected to maintain tight credit underwriting standards in the second quarter of 2020[6/], given deteriorating loan qualities of business loans (both loans extended to large businesses and SMEs), credit card loans, and auto leasing. However, the government's soft loan programs would help expand accessibility to credits for the private sector and somewhat facilitate more loan extensions by financial institutions.

The baht against the U.S. dollar and the nominal effective exchange rate (NEER) appreciated throughout the second quarter due to the relaxation of containment measures in the region and sales of U.S. dollars by domestic players.

In the second quarter of 2020, **the baht against the U.S. dollar** appreciated relative to the end of the previous quarter (Chart 2.16). At the beginning of the quarter, the baht weakened in light of high uncertainties surrounding the global spread of the COVID-19 in the previous quarter, as well as U.S. dollar strength during the period of dollar liquidity strain. However, the baht subsequently strengthen in line with other regional currencies. This was because many countries, especially Asian countries including Thailand, were successful in containing the spread of the virus, allowing governments to gradually relax containment measures and economic activities to slowly pick up. At the same time, accommodative monetary and fiscal policies implemented by major advanced economies helped boost investor confidence, causing capital outflows by non-residents to slow and inducing the return of capital into regional markets in June, especially into the bond markets. Nevertheless, the strength of the baht was also supported by domestic factors including sales of U.S. dollars associated with the sales of gold when pandemic uncertainties drove up the gold price, as well as sales of U.S. dollar forward contracts by institutional investors that invested abroad in order to hedge against foreign exchange risk. Thus, the baht averaged at 31.22 baht to U.S. dollar in June 2020, a 2.8 percent appreciation from an average of March 2020.

Chart 2.16 The baht appreciated against the U.S. dollar following the relaxation of containment measures in regional countries and foreign currency sales by domestic players

USDTHB, NEER, DXY

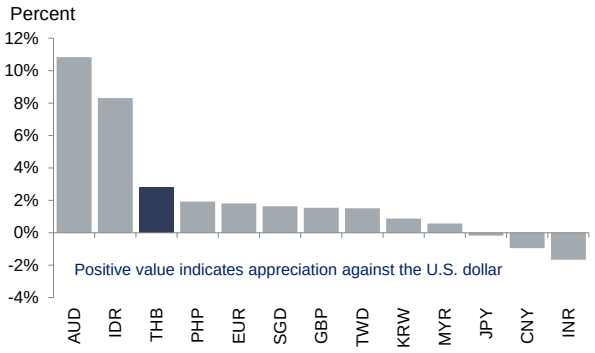


Source: Bank of Thailand, Bloomberg and Reuters (data as of June 23, 2020)

[6/] Results of financial institutions' survey as published in the Report on Credit Conditions Q1/20 and Outlook for Q2/2020.

The Nominal Effective Exchange Rate (NEER) index averaged at 123.88 in June 2020 (as of June 23, 2020), 1.8 percent higher than the average of March 2020. This was attributable to the baht appreciation against most trading partner currencies (Chart 2.17). Meanwhile, the preliminary **Real Effective Exchange Rate (REER)** in May 2020 weakened 1.2 percent from an average of March 2020, mainly due to a larger fall in Thailand's consumer price index relative to those of trading partner countries.

Chart 2.17 The baht appreciated against the U.S. dollar, in line with movements of most regional currencies.
(Average in June 2020 compared to average in March 2020)



Source: Bank of Thailand and Reuters (data as of June 23, 2020)

Going forward, exchange rates would likely remain volatile pertaining to risk of the second wave of the outbreak, monetary and fiscal policies in major advanced and regional economies that would remain accommodative, and high uncertainties surrounding the recovery of trading partner economies. In addition, global financial stability risk from corporate debt defaults in many countries, including Thailand, could result in sharp and sudden corrections in global asset prices. Against such backdrop, the Committee expressed concerns over the baht appreciation which could affect the economic recovery. Therefore, the Committee deemed it necessary to closely monitor developments in the foreign exchange markets in order to assess the necessity of implementing additional appropriate measures.

The Thai financial system became more vulnerable due to this year's contraction of the economic outlook in light of the COVID-19 situation. Vulnerabilities increased across multiple fronts, especially the deterioration in debt servicing ability of businesses and households.

The Thai financial system became more vulnerable due to this year's contraction of the economic outlook in light of the COVID-19 pandemic and the impact of the containment measures on household and business income. Nevertheless, fiscal sector stability remained sound, providing fiscal policy space to support the economic restoration and restructuring in the coming periods. Nevertheless, debt-to-GDP ratio would likely increase, but would remain at a manageable level. Amid high uncertainties surrounding the COVID-19 situation and risk of a larger-than-expected contraction in the global economy, risks to financial stability that warranted monitoring could be summarized as follows.

1. Although volatilities in the Thai financial markets subsided after the introduction of measures to stabilize financial markets, there remained a need to closely monitor risks in the financial markets going forward. The BOT, in collaboration with relevant agencies, implemented a series of measures to stabilize financial markets as preventive measures against systemic risks. These measures ranged from outright purchases of government bonds to enhance liquidity in the bond market, the establishment of the MFLF and the BSF, as well as the policy rate cuts. These measures helped boost confidence of investors, resulting a slower redemption of their investment in the bond market. Financial market volatilities thus declined and effective market functioning was restored. Furthermore, risk of the U.S. dollar liquidity squeeze in the Thai financial markets remained low, thanks to the high

level of international reserves. Also, external debt to GDP was low and the current account remained in surplus, suggesting a modest demand for U.S. dollar in the short term.

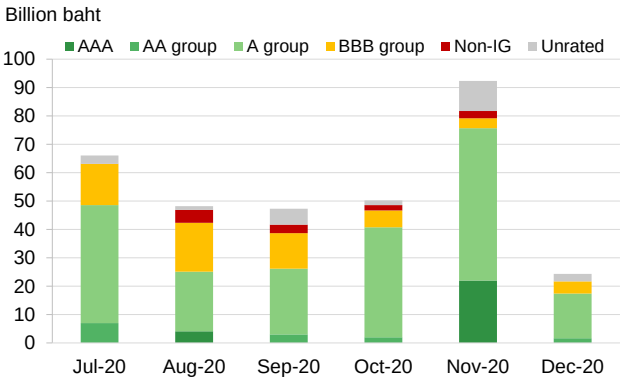
Nevertheless, there remained a need to closely monitor risks in the financial markets, given high uncertainties pertaining to the global and Thai economic recovery, as well as financial market developments. Should investors turned more pessimistic about the severity of the global economic contraction, this could trigger sharp corrections in global asset prices or corporate debt defaults in many countries, which could in turn affect the Thai financial markets.

2. Liquidity and debt servicing ability of households and businesses deteriorated in light of worsening economic conditions. **Household financial positions became more fragile with rising risk of defaults.** This was reflected in the household debt to GDP ratio, which remained elevated at 80.1 percent in the first quarter of 2020. Also, the NPL ratio of consumer loans rose from 2.9 percent in the previous quarter to 3.2 percent in the first quarter of 2020[7/], in line with deteriorating quality of consumer loans across all loan categories, especially credit card and personal loans. Nonetheless, households with monthly income of less than 15,000 baht had risk of liquidity shortages and low buffer against income shocks arising from the COVID-19 pandemic. Since nearly all of their income had to be spent on consumption expenses, their savings would be insufficient to provide cushion against severe income shocks. If these households were to lose their income, their savings would only cover their expenses for no more than 3 months[8/].

Similarly, **financial positions of businesses became more vulnerable with higher risk of defaults**. This was due to many factors. First, the outlook of corporate income worsened in light of the COVID-19 pandemic, as indicated by the operating profit margin (OPM) and interest coverage ratio (ICR) which remained persistently in the negative territory[9/]. In particularly, businesses that warranted close monitoring included those experiencing severe impacts and unable to recover quickly, such as hotels and restaurants. In addition, some businesses were already financially fragile prior to the pandemic due to elevated debt levels, such as businesses in real estate and construction, or due to weak profitability, such as SMEs in textile and trading sectors. Second, investors in the corporate bond markets had lower risk tolerance, leading to rollover risks among issuing companies. Furthermore, some businesses that suffered impacts from the COVID-19 could face risk of credit-rating downgrades to non-investment grade,

Chart 2.18 Some businesses affected by impacts of the COVID-19 might face credit-rating downgrades though most corporate bonds maturing in 2020 were rated A and BBB.

Outstanding value of corporate bonds nearly matured



Source: ThaiBMA, compiled by Bank of Thailand

[7/] Deteriorating quality of consumer loans in the first quarter of 2020 was partly a result of adjustments regarding classification of financial instruments under the Thai Financial Reporting Standards 9 (TFRS 9), which was adopted for the first time in this quarter.

[8/] The calculation was based on the Socio-Economic Survey (SES) in 2019.

[9/] As reflected by the ICR and OPM of 25th-percentile small-sized listed companies, which remained in the negative territory.

although most corporate bonds maturing in 2020 were rated A and BBB (Chart 2.18). Moreover, corporate bond default risks could trigger knock-on effects to other related parties in the financial system. These included, for example, cross-default, especially in the case that creditors were commercial banks, and savings cooperatives with significant holdings of corporate bonds, as well as mutual fund unit holders.

In recent periods, the government and the BOT had implemented a series of fiscal, financial, and credit measures to alleviate the impact of the COVID-19 on businesses and households (**BOX**: *Recent progress on the Bank of Thailand's financial and credit relief measures to address the impact of COVID-19*).

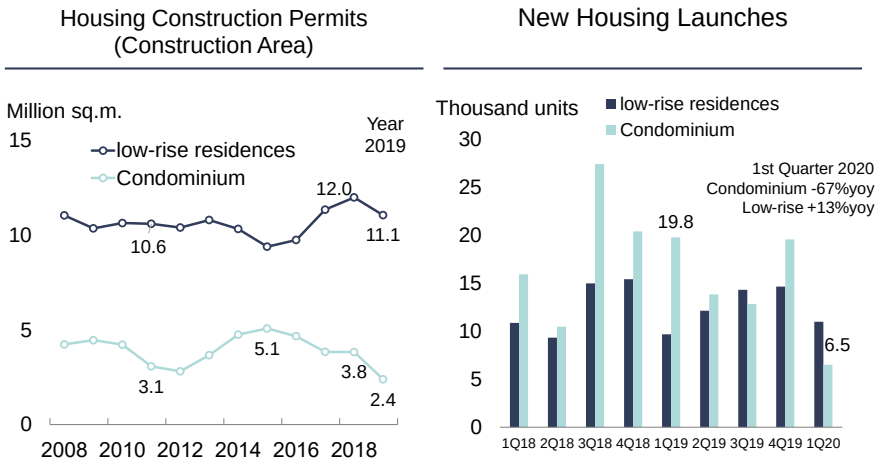
3. Although stability of financial institutions remained robust, it warranted close monitoring given deteriorating debt servicing ability of businesses and households amid high uncertainties surrounding the spread of COVID-19. Stability of financial institutions remained sound, as reflected by adequately high levels of commercial banks' capital buffers and reserves, providing cushion against risks of deteriorating credit quality. At the end of the first quarter, capital buffers in the commercial banks' system totaled 2,836 billion baht, constituting a BIS ratio of 18.7 percent. Reserves stood at 719.2 billion baht, leading to a reserves-to-NPL ratio of 143.3 percent.

Maintaining a high level of capital was considered an important measure to ensure commercial banks' ability to withstand future uncertainties and preserve the lending capacity needed to support the economic recovery after the COVID-19 pandemic subsided. As a result, on June 18, 2020, the BOT issued a circular asking commercial banks to suspend interim dividend payments for the year 2020, and share buybacks, early redemptions or purchases of tier-1 or tier-2 capital securities[10/]. This would help maintain strong capital positions of commercial banks in the coming periods.

4. The real estate market still faced high risk from falling demand for residences from domestic and foreign buyers, especially during the first half of 2020 due to the COVID-19 pandemic. The oversupply was thus expected to rise. In light of the situation, some developers began to make some adjustments regarding their supplies in the first quarter of 2020, as reflected in the 67 percent reduction in new condominium launches compared to the same period last year (Chart 2.19). However, there remained certain behaviors that warranted monitoring as they could lead to the build-up of vulnerabilities. In particular, some developers engaged in competition to reduce excess supplies, such as agreeing to cover mortgage installments for their buyers for the period of 1-5 years. These mortgages would be at risk of becoming NPL in the future, if buyers could not afford the installments once supports from developers ended. In addition, financial positions of some developers that relied on funding through the corporate bond market should also be monitored given risks from higher funding costs and credit-rating downgrades.

[10/] BOT Circular No. BOT.FPG (23)c 643/2020 The enhancement of commercial banks' capital buffers to withstand the risks from the coronavirus pandemic (June 18, 2020).

Chart 2.19 Developers slowed down new project launches in tandem with contracting demand. However, competitions in sales promotions should be monitored as they could contribute to build-up of risks



Source: REIC, AREA calculated by Bank of Thailand

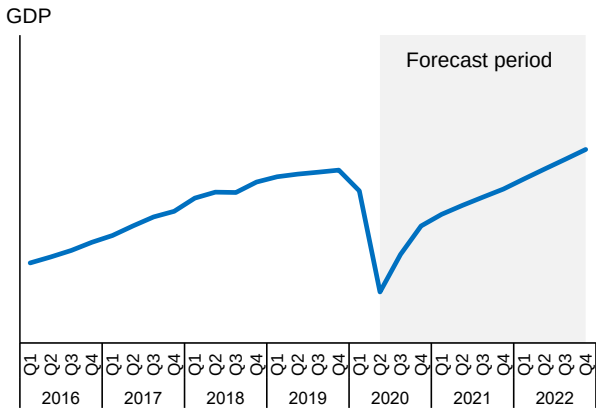
2.2 Outlook for the Thai Economy

The Thai economy would contract in 2020 more than previously assessed in the previous *Monetary Policy Report*. Largest contraction was expected in the second quarter due to a-more-severe-than-expected COVID-19 pandemic and the economy would gradually recover from the third quarter onward. Meanwhile, headline inflation would be more negative, mainly on account of energy prices.

The Committee assessed that the Thai economy would contract 8.1 percent in 2020, more than previously assessed in the previous *Monetary Policy Report*. Such severe contraction would be observed across almost all economic growth components, except public expenditure. This was due to a more-severe-than-expected COVID-19 pandemic and the containment measures implemented by governments in many countries including Thailand. This led to temporary disruptions of global economic activities especially in the second quarter of 2020. Thailand's tourism and merchandise exports were thus projected to markedly contract, which would weigh on private consumption as a result of weakening income and employment. In addition, economic activities would recover slower than expected, as people were still concerned over the outbreak despite the government's success in containing the spread of the virus in the second quarter of 2020. Furthermore, weaker domestic and external demand would result in a sharp decline of private investment. However, public expenditure, together with transfers, tax, and other financial measures, would play an important role in shoring up the economy in the period ahead. Nevertheless, **the Thai economy was projected to bottom out in the second quarter of this year and would gradually recover from the third quarter onward,** (Chart 2.20) following the relaxation of containment measures in many countries, including Thailand, and the re-opening of the country for inbound foreign tourists under the travel bubble scheme. Merchandise exports, tourism, and domestic demand would therefore begin to improve. **In 2021, the Thai economy**

was projected to expand at 5.0 percent, higher than previously assessed in the previous *Monetary Policy Report*. This was partly due to the lower base of tourism and merchandise exports in 2020, and as government was exp ected to launch additional employment stimulus packages and large-scale investment projects. However, the Thai economy would still remain substantially below its potential level given changing patterns of business operations, way of life, and employment patterns in the post-COVID environment. This could cause some businesses to not fully recover.

Chart 2.20 The Thai economy is expected to recover from Q3/2020 onward



Additional assumptions underlying the economic projection in this *Report* included, first, there would be no severe second-wave of the COVID-19 outbreak in Thailand. This would allow for the gradual re-opening of the country under the travel bubble scheme, which would be done through partnerships between COVID-19-free countries, permitting travelling into targeted locations in each countries starting from the second half of 2020. Second, there would be a widespread availability of COVID-19 vaccines from the fourth quarter of 2021 onward, despite delaying from previously anticipated from mid-2021.

Headline inflation in 2020 would be more negative than previously assessed, mainly due to supply-side factors, where it was projected to be negative at 1.7 percent, before gradually rising to 0.9 percent in 2021. Headline inflation would return to the target in 2021 in line with gradually increasing crude oil prices and the economic recovery.

Table 2.2 Forecast summary

Percent	2019*	2020	2021
GDP growth	2.4	-8.1 (-5.3)	5.0 (3.0)
Headline inflation	0.7	-1.7 (-1.0)	0.9 (0.3)
Core inflation	0.5	0.0 (-0.1)	0.1 (0.1)

Note: * Outturn

( ) Monetary Policy Report March 2020

Sources: NESDB, Ministry of Commerce, Bank of Thailand's estimates

Summary of the key forecast assumptions

- Trading partner economies were projected to markedly contract in 2020 due to the COVID-19 outbreak that became more severe, widespread, and prolonged than expected. This led to larger-than-expected disruptions to global economic activities in the second quarter. However, trading partner economies were expected to recover slowly from the third quarter of 2020 onward following the relaxation of lockdown measures and a series of additional stimulus packages implemented by governments in many countries, including

both financial and fiscal measures. Trading partner economies would thus grow at a high rate in 2021, mainly due to the low base effect, but overall economic activities would remain below the pre-pandemic level.

- The Federal funds rate was projected to be maintained at a range of 0.00 - 0.25 percent throughout the forecast horizon.
- Regional currencies (excluding the Chinese yuan) depreciated slightly more than previously projected throughout the forecast horizon due to a larger-than-expected impact of COVID-19 pandemic on Asian economies. This was reflected in a weaker-than-assessed regional currencies index. However, regional currencies would gradually strengthen in the period ahead in line with a better recovery of Asian economies over major advanced economies after the pandemic subsided. In addition, the U.S. dollar was expected to weaken amid concerns over the political protest situation and prolonged outbreak in the U.S., as well as the outlook of trade tensions between the U.S. and China which could intensify prior to the U.S. presidential election in November.
- Dubai crude oil prices would remain largely unchanged in 2020 and would gradually rise in the following year in line with gradually recovering global economy and a larger-than-expected production cut, following an agreement to cut crude oil production of major oil producers and crude oil production cut by the U.S. as the revenue was not sufficient to compensate the costs. However, crude oil prices would remain at low levels throughout the forecast horizon.
- Farm income (excluding government support measures) was expected to contract by a larger extent in 2020 due to both output and prices. Agricultural output would decline more than previously assessed in line with lower volume of rice and sugarcane due to a larger-than-expected impact of the drought. Meanwhile, agricultural prices were expected to fall mainly in line with prices of rubber as a result of a large drop in external demand, although prices of white rice rose from the stocking-up of food during the outbreak. In addition, farm income in 2021 would largely expand due to the low base effect, as well as better assumptions on the weather and dam water levels toward the end of the rainy season in 2020.
- The Emergency Decree Authorizing the Ministry of Finance to Raise Loans to Solve Problems, to Remedy and Restore the Economy and Society as Affected by the Coronavirus Disease Pandemic in 2019 (B.E. 2563) totaling 1 trillion baht and the reallocation of the budget structure for a faster disbursement to remedy and restore the economy thanks to the Budget Expenditure Transfer Bill B.E. 2563 (A.D. 2020) and the revision of the fiscal year 2021 budget structure.

Table: Summary of forecast assumptions

Annual percentage change	2019*	2020	2021
Dubai crude oil price (U.S. dollar per barrel)	63.5	35.1 (35.0)	37.0 (34.0)
Farm income (% YoY)	3.3	-3.9 (-2.5)	6.2 (5.0)
Government consumption at current price (billion baht)[1/]	2,723	2,871 (2,863)	3,028 (2,993)
Public investment at current price (billion baht)[1/]	966	1,029 (1,040)	1,184 (1,141)
Fed funds rate (% at year end)	1.50 - 1.75	0.00 - 0.25 (0.00 - 0.25)	0.00 - 0.25 (0.00 - 0.25)
Trading partners' GDP growth (% YoY)[2/]	2.7	-4.6 (0.2)	4.2 (2.0)
Regional currencies (excl. China) vis-à-vis the U.S. dollar (index)3/	156.6	159.5 (158.7)	159.5 (158.7)

Notes: [1/] Assumption includes spending on infrastructure investment plans
[2/] Weighted by each trading partner's share in Thailand's total exports
[3/] Increasing index represents depreciation, decreasing index represents appreciation
* Outturns
( ) *Monetary Policy Report* March 2020

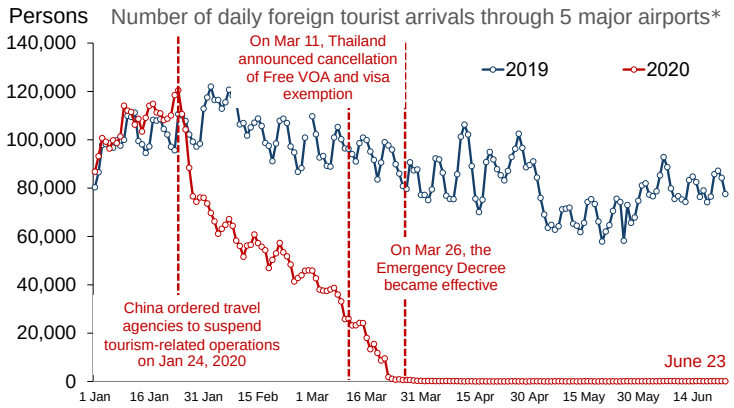
Merchandise exports would exhibit a more-severe contraction than the previous assessment.

The value of Thai merchandise exports in 2020 would experience a sharper decline than previously assessed, due mainly to the contraction in volume. This was in line with the outlook of trading partner economies and global trade volume that sharply contracted, as well as supply chain disruptions in some businesses. Export indicators suggested that Thai exports would continue to markedly contract in the second quarter. Nevertheless, the gradual relaxation of containment measures in many countries would support merchandise exports to improve in line with economic activities. Meanwhile, prices of merchandise exports were projected to decline, mainly owing to prices of goods that were sensitive to movements of crude oil prices. **In 2021, the value of merchandise exports was expected to expand,** both in terms of volume in response to the recovery of global economy and global trade, and in terms of price in line with the outlook of global crude oil prices. However, the value of merchandise exports would be substantially below the pre-pandemic level, as changing patterns of business operations under the new way of life in the post-COVID environment would limit the potential of certain industries to fully recover. **The Committee therefore assessed the value of merchandise exports to sharply contract by 10.3 percent in 2020 before expanding at 4.5 percent in 2021.**

Exports of services were expected to contract more severely in line with the number of foreign tourists.

The tourism sector was projected to recover more slowly than the previous assessment, (Chart 2.21) although the government was successful in containing the spread of the virus within the second quarter of 2020 as previously assessed. However, the outbreak in other countries would be more prolonged and widespread than previously expected. The Thai government was thus expected to maintain its travel restrictions on inbound foreign tourists for an extended period and would begin to gradually re-open the country for foreign tourists under the travel bubble scheme in the second half of 2020. **The foreign tourist figures were therefore projected to be 8.0 and 16.2 million persons in 2020 and 2021, respectively, lower than previously assessed in the previous *Monetary Policy Report.***

Chart 2.21 No foreign tourists was expected to visit Thailand in Q2/2020



Note: *5 main airports include Suvarnabhumi, Don Mueang, Chiang Mai, Hat Yai and Phuket International Airports, accounting approximately 83% of total foreign tourists.

Source: TAT Intelligence, Ministry of Tourism and Sports compiled by BOT

The current account balance was projected to register a surplus of 15.5 and 20.2 billion U.S. dollars in 2020 and 2021, respectively. The current account balance in 2020 was revised down from the previous assessment due to a larger deficit in the services balance, mainly as a result of a marked decline in tourism receipts. Meanwhile, trade balance was expected to record a slightly lower surplus, although the value of merchandise exports largely declined. This was as the value of imports also experienced a sharp reduction in response to lower demand for imports of both raw materials and intermediate goods for export-related manufacturing and domestic sales. In 2021, the current account balance was projected to increase from the previous year, owing to gradual improvements in trade balance and tourism receipts.

Public expenditure would be key economic growth driver and would exhibit a higher-than-expected growth than previously assessed.

Public expenditure would play an important role in shoring up the economy in 2020 and supporting the economic recovery in the coming periods. The government was expected to accelerate disbursement of the fiscal year 2020 budget during the rest of the year, as well as reallocate the budget structure for a faster and continuous disbursement to remedy and restore the economy thanks to the Budget Expenditure Transfer Act, B.E. 2563 (A.D. 2020) and the revision of the fiscal year 2021 budget structure. There would also be an additional liquidity support from the Emergency Decree Authorizing the Ministry of Finance to Raise Loans to Solve Problems, to Remedy and Restore the Economy totaling 1 trillion baht, which was part of the third phase of government relief measures to alleviate the impact of COVID-19. The liquidity support included (1) spending on public health, (2) transfers to remedy those affected by the outbreak under projects, such as “You will never be left behind” project and project to remedy farmers and vulnerable groups, and (3) spending to restore the economy and society in the coming periods. In addition, the government would introduce further measures to help those affected by the outbreak, including credit measures through specialized financial institutions (SFIs) and tax measures. This would partly help sustain business operations and employment. Nevertheless, investment spending by state-owned enterprises was projected to expand, despite slowing down somewhat from the previous assessment due to adjustment in investment plan in response to the COVID-19 situation.

Private consumption would contract more than previously assessed, but started to show signs of a gradual improvement in the period ahead.

Private consumption would contract in 2020 more than previously assessed, especially in the second quarter where severe contraction would be observed due to the impact of containment measures, as well as weakening non-farm income and consumer confidence. However, domestic outbreak situation that began to be contained would allow the government to gradually relax containment measures and facilitate a gradual pick-up of economic activities in the period ahead. Nevertheless, a series of government measures aimed at supporting those affected by the outbreak and boosting domestic tourism in the second half of this year would help boost non-farm income and private consumption to some extent. Meanwhile, farm income was expected to gradually recover as the drought situation improved. Nevertheless, private consumption would be below the pre-pandemic level throughout the forecast horizon, as households would be more cautious in their spending in the period ahead amid heightening

uncertainties on the COVID-19 situation. Moreover, households' financial positions became more fragile given declining income and elevated debt levels.

Private investment was expected to contract in the short run in line with lower domestic and external demand

Private investment was expected to contract in 2020 more than the previous assessment. This was on account of the contraction in both domestic and external demand, as well as high levels of excess production capacity, which caused many businesses to alter patterns of business operations. Coupled with the uncertain COVID-19 situation, most businesses thus postponed investment plans. However, government's infrastructure investment projects were underway, particularly the public-private partnership (PPP) projects both within and outside the Eastern Economic Corridor (EEC). **As for 2021, some investment projects would likely continue**, particularly large-scale infrastructure investment and infrastructure investment to support the digital economy, including accelerated investment in the 5G network following the successful auction in the first quarter of 2020. In addition, the COVID-19 pandemic resulted in higher demand from households and businesses for technology products. Furthermore, merchandise exports and tourism would gradually recover which would partly support the expansion of private investment.

Projection of headline inflation would be more negative in the short term due to supply-side factors

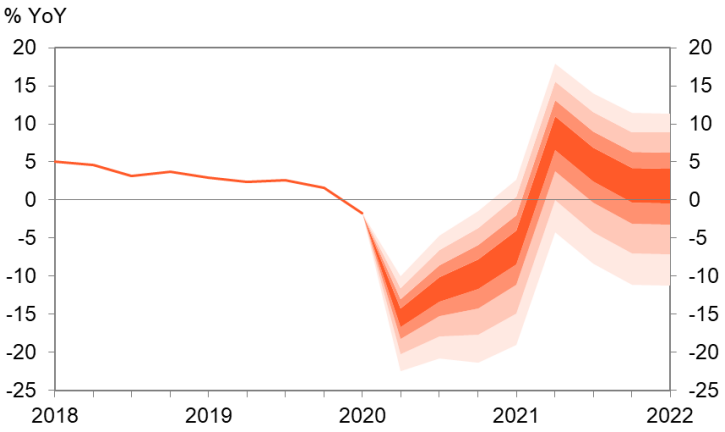
Projection of headline inflation in 2020 would be negative at 1.7 percent, lower than previously assessed in the previous *Monetary Policy Report*, mainly owing to supply-side factors. This was particularly because of sharply declined energy prices in line with global crude oil prices, which was owing to decreasing demand following disruptions of global economic activities and high levels of excess production capacity. Moreover, a fall in headline inflation was partly due to government measures to reduce electrical and water bills. Meanwhile, **core inflation would remain low** in line with muted demand-pull inflationary pressures. **As for 2021, headline inflation was expected to gradually rise to 0.9 percent and return to the target** mainly in tandem with gradually rising crude oil prices. **Core inflation was projected to be zero and 0.1 percent in 2020 and 2021, respectively,** in response to gradual economic recovery outlook.

Economic and inflation projections were subject to greater downside risks and were highly uncertain

The Committee assessed that the Thai economy in the period ahead would be highly uncertain, subject to downside risk, and mainly depended on the COVID-19 impact. This was reflected in a wider fan chart that tilted toward the downside. (Chart 2.22) **The possibilities that the Thai economy would underperform the baseline projection would be due to the following factors.** First, the global economy could recover more slowly than expected or enter a severe recession on account of several factors including an emergence of the second wave of the outbreak in many countries, which would lead to larger-than-expected impact on Thai merchandise exports. Second, default risks of businesses and households could increase. Third, the recovery of the tourism sector could be slower than expected including a slower-than-expected process of the travel bubble scheme. Forth, Thailand could experience the second wave of the COVID-19 outbreak, triggering

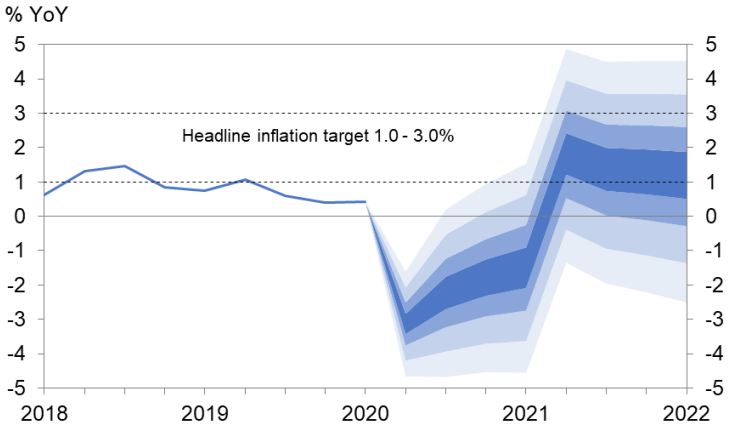
another round of lockdown measures. However, **there were possibilities that the Thai economy would outperform the baseline projection due to** (1) better-than-expected impacts of additional government's stimulus measures to shore up domestic consumption and investment, and measures to assist borrowers, (2) public infrastructure investment and PPP projects which could incentivize the private sector to invest more than expected, and (3) more relocations of production base to Thailand thanks to Thailand's success in containing the spread of the virus. Meanwhile, **there remained downside risks to headline and core inflation projections** (Chart 2.23 and 2.24) in line with risks to crude oil prices and the economic growth projection.

Chart 2.22 Growth forecast



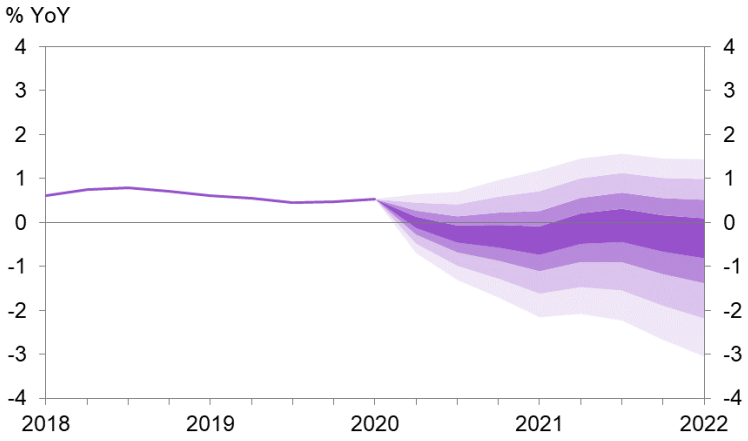
Note: Fan chart covers 90% of the probability distribution

Chart 2.23 Headline inflation forecast



Note: Fan chart covers 90% of the probability distribution

Chart 2.24 Core inflation forecast



Note: Fan chart covers 90% of the probability distribution

Table 2.3 Forecasts of GDP and components

Annual percentage change	2019*	2020	2021
GDP growth	2.4	-8.1 (-5.3)	5.0 (3.0)
Domestic demand	3.3	-3.4 (-0.8)	4.0 (2.6)
Private consumption	4.5	-3.6 (-1.5)	2.5 (2.1)
Private investment	2.8	-13.0 (-4.3)	5.6 (2.2)
Government consumption	1.4	3.8 (2.6)	3.1 (2.2)
Public investment	0.2	5.8 (5.8)	14.1 (8.1)
Exports of goods and services	-2.6	-22.7 (-16.4)	8.4 (5.0)
imports of goods and services	-4.4	-18.9 (-11.6)	7.3 (4.3)
Current account (billion, U.S. dollars)	38.4	15.5 (19.4)	20.2 (19.2)
Value of merchandise exports	-3.3	-10.3 (-8.8)	4.5 (0.2)
Value of merchandise imports	-5.6	-16.2 (-15.0)	4.1 (-0.4)
Number of foreign tourists (million person)	39.8	8.0 (15.0)	16.2 (20.0)

Note: *Outturns

( ) *Monetary Policy Report* March 2020

Assessing risk of deflation in Thailand

The severe COVID-19 pandemic during the first half of 2020 prompted governments in many countries, including Thailand, to implement containment measures, which led to unprecedented disruptions of global economic activities. Moreover, the changing global and Thai economic environment substantially affected Thailand's inflation in the recent periods, and partly explained the negative inflation which averaged at -1.04 percent during the first five months of 2020. Inflation has remained in the negative territory since March and reached the lowest level in a decade at –3.44 percent in May. This led to the question of whether Thailand would experience deflation. The Committee thus saw the need to develop a better understanding among the public regarding the definition, the causes, and impacts of deflation, as well as to assess whether Thailand is currently experiencing deflation.

Definition, Causes, and Impacts of Deflation

Inflation is narrowly defined as a situation when overall price levels of goods and services experience a persistent contraction. For this narrow definition, however, the assessment criteria adopted by each international organization varies. For instance, the ECB[11/] set the criteria at three consecutive months of negative inflation, while the IMF[12/] used the criteria of 3 - 6 consecutive months. Deflation could arise as a result of **supply-side factors**, when price levels of goods and services fell as a result of lower costs of production thanks to higher labor productivity, better production technology, or cheaper raw materials. Lower prices as a result of these supply-side factors help stimulate consumer demand and will, in turn, benefit the overall economy. On the other hand, deflation could be caused by **demand-side factors**, whereby prices of goods and services fall as a result of subdued demand. This typically occurs during a prolonged economic recession where consumers lack purchasing power, or expect prices to fall further in the future and therefore postpone their spending. Producers then need to cut prices in order to stimulate sales, as well as reduce both production and investment. This consequently affects employment and labor income, which will further depress the economy and drive down prices, resulting in a deflationary spiral.

However, the assessment of deflation according to the narrow definition above only takes into account the period of time that overall prices of goods and services experience a contraction. **An appropriate assessment of deflation should instead follow a broader definition which takes into consideration the causes and impacts of deflation in a comprehensive manner.** Deflation, in its broad definition, occurs when prices of goods and services experience a persistent contraction that are broad-based across most categories, and the central bank is unable to anchor inflation expectations of the public, which mostly occurs when demand and employment face a prolonged contraction. According to this broad definition, countries that used to experience deflation include the U.S., many countries in the euro area. Deflation was also observed in Japan during the 1930s, and from the end of 1990s until the beginning of 2000s.

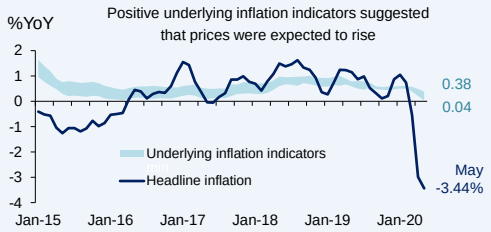
[11/] European Central Bank (2014), "Risk of Deflation." Monthly Bulletin, June.
[12/] Kumar et al. ((2003, "Deflation: Determinants, Risks, and Policy Options." IMF Occasional Paper No 221

Current risk of deflation in Thailand

Negative headline inflation that occurred in Thailand for three consecutive months since March 2020 was not a sign of deflation, as the situation did not meet all four criteria according to a broad definition of deflation. (Chart 1) Details are as follows.

Chart 1 Headline inflation that had been in the negative territory for many months was not a sign of delation, as the situation did not meet all four criteria according to a broad definition

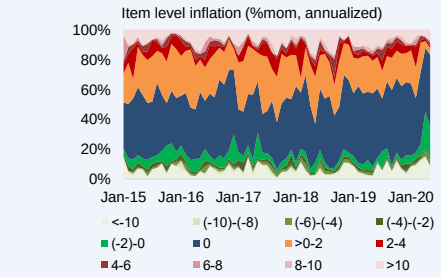
Criteria 1: Overall prices of good and services decline continuously



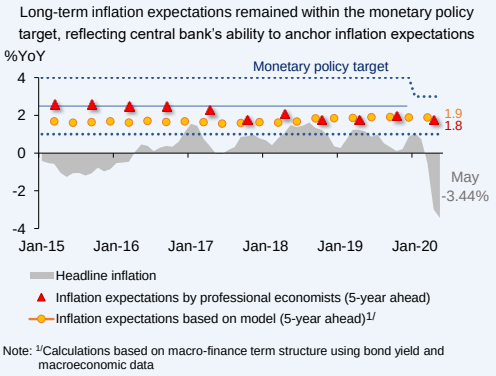
Note: Shaded area show the highest and lowest outcome among different measures of underlying inflation. The measure included are as follows.
(1) **Asymmetric trim** excluding goods and services with most volatile price changes, removing the bottom 15 percentile and the top 10 percentile
(2) **Principal component model** and
(3) **Core inflation excluding rents and government measures**

Criteria 2: The decline in prices of goods and services can be observed across many product categories

Shares of items in the CPI basket were based on price adjustments from the previous month, which reflected a stable or improvement for most prices



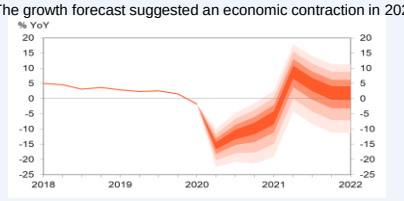
Criteria 3: Inflation expectations fall below the monetary policy target



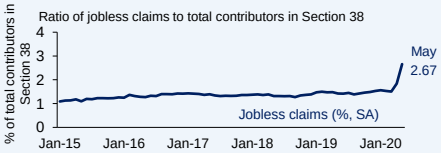
Note: 1/Calculations based on macro-finance term structure using bond yield and macroeconomic data

Criteria 4: Demand and employment become persistently subdued.

The growth forecast suggested an economic contraction in 2020



In May, the number of jobless claims accelerated



Source: The Ministry of Commerce, Social Security Office, Consensus Economics, calculations by BOT

Criteria 1: a persistent contraction in the prices of goods and services

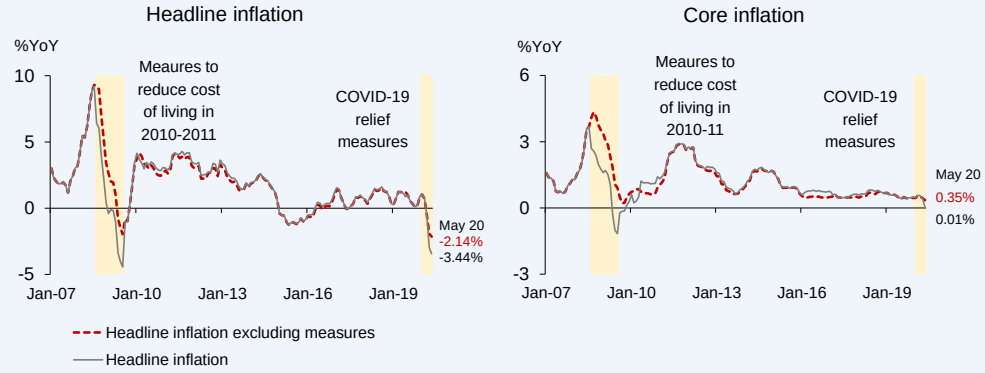
Persistent negative headline inflation in Thailand during the first half of 2020 was mainly a result of supply-side factors. These included, in particular, a sharp fall in energy prices. This was as domestic retail oil prices fell in line with global crude oil prices and the government measures to reduce electricity charges to alleviate the impact of the COVID-19 outbreak. However, considering the underlying inflation which is used to assess future's inflation suggested that **headline inflation would rise in the period ahead, in line with the BOT's latest projection which expected headline inflation to turn positive in 2021**.

Criteria 2: a broad-based contraction in the prices of goods and services in the inflation basket

The prices of goods and services in Thailand's inflation basket contracted only in some categories. Prices of around 70 percent of goods and services in the inflation basket increased or remained stable. Meanwhile, the number of goods which experienced falling prices increased somewhat in recent periods. However, this was, in part, the result of government

me asu res to alleviate the impact of the COVID-19 (Chart 2), such as reducing electricity and water bills for three months. In addition, the government has also requested cooperation from businesses to reduce costs of living for households, thereby leading businesses to launch more promotions.

Chart 2 Government measures to mitigate impacts of the COVID-19 partly led to a decline in inflation



Note: Government measures include measures to reduce expenses related to electricity, LPG, water supply, transportation, education and taxes.

Source: The Ministry of Commerce, calculations by BOT

Criteria 3: the central bank is unable to anchor inflation expectations

Long-term headline inflation expectations remained within the monetary policy target. According to the most recent survey of professional forecasters in April 2020, long-term (five-year ahead) headline inflation expectation was at 1.8 percent and therefore remained within the target range of 1.0 – 3.0 percent. This reflected the ability to **anchor inflation expectations, although headline inflation** continues to decline and has been below the lower bound of the target during the past months.

Criteria 4: Persistent decline in demand and employment

Following the COVID-19 outbreak, **economic activities in Thailand dampened significantly due to the implementation of containment measures and a slowdown in trading partner economies.** This directly weighed on employment and household income, which then translated to deteriorating purchasing power and a contraction in domestic demand. Coupled with the outlook of a slowly recovering global economy, this then led to excess production capacity in many industries, and therefore resulted in low price levels of goods and services. **Nonetheless, the Committee saw a need to monitor and assess whether demand and employment would become persistently subdued** after the relaxation of the containment measures.

In summary, the current economic development in Thailand did not exhibit signs of deflation according to its broad definition. **Going forward, deflation risk could rise, should the global as well as the Thai economies face a more-severe contraction or a much slower recovery than expected. The Committee would thus monitor developments of the economy and inflation closely**, particularly factors which could affect consumption and private investment. These include both income and employment, as well as household and

business confidence. Moreover, the inflation outlook in the period ahead would be highly uncertain due to structural changes in the global economy after the COVID-19 outbreak, trade tensions and geopolitical risks in many parts of the world. These could potentially lead to deglobalization, which could have significant impacts on the global supply chains and prices of goods. The Committee therefore placed importance on clear and continuous communication with the public regarding the outlook of inflation and risk of deflation.

Recent progress on the Bank of Thailand's financial and credit relief measures to relieve impacts of the COVID-19

In recent periods, the Bank of Thailand (BOT) has been implementing a series of measures to address liquidity problems facing households and businesses affected by the COVID-19 pandemic. Given high degree of uncertainties surrounding the pandemic, particularly how and when it would end, households and businesses with liquidity problems could thus face solvency risks, which would hinder economic recovery after the pandemic. In implementing the financial relief measures amidst high uncertainties, the BOT aims to address such problems in accordance with the following key principles.

1. The supports must reach those affected by the pandemic in a broad-based and timely manner.
2. They must not pose risks to the financial institution system.
3. They must not cause excessive fiscal burden on the government or tax burden on taxpayers going forward.
4. They should support the transformation of businesses to prepare for changes in consumer behaviors and business operations in the post-COVID environment.

Recent progress on four financial measures implemented thus far can be summarized as follows.

(1) Interest rate cuts Since February 2020, the Monetary Policy Committee (MPC) has delivered three policy rate cuts, and reduced the Financial Institutions Development Fund (FDIF) contribution rate to 0.23 percent of the deposit base for the duration of 2 years. These moves enabled financial institutions to cut the reference loan rates to the lowest level in 16 years, which helped lower the interest burden for floating-rate loans.

In addition, the BOT asked financial institutions to offer a 2 - 4 percent reduction in interest rate ceilings, service fees, and penalty charges on fixed-rate credit card and personal loans, effective from on August 1, 2020 onwards. Moreover, an adjustment was made to the calculation of late payment fees. Under the new calculation method, interest charges will only be applied on the specific installment that was missed, instead of on the entire outstanding loan. This new method would reduce interest burden on borrowers, and reflect the actual loss to the lenders.

(2) Debt payment holiday and debt reduction Empowered by the Emergency Decree on the Provision of Financial Assistance for Entrepreneurs Affected by the COVID-19 Pandemic, B.E. 2563 (2020) (Financial Assistance Emergency Decree), **the BOT issued a loan payment holiday of 6 months for all SMEs** with normal repayment status and a credit line not exceeding 100 million baht per financial institution as of end-2019. During the payment holiday, the pause payments will not be considered as missed payment. Furthermore, the BOT engaged with the Thai Bankers' Association to encourage commercial banks to consider **additional liquidity supports for SMEs** beyond measures previously announced.

For retail borrowers, including small SMEs who rely on personal loans, auto loans, and leasing for working capital to support business operations, on March 26, 2020, the BOT, together with financial institutions and non-bank financial services providers, agreed to adopt **minimum measures on debt moratorium and debt reduction to assist retail borrowers of all loan types for the duration of 3 months**. During this period, payment suspensions

would not be considered as missed payments. These measures would ensure sufficient cash flows for retail borrowers to sustain necessary expenditures and continue their business operations. Furthermore, financial institutions and other financial services providers continued to offer additional measures that were better than the minimum measures. As these phase I measures will soon expire, while the COVID-19 situation remains highly uncertain, **the BOT thus announced the Phase II minimum measures to assist retail borrowers** on June 19, 2020 to expand the scope and duration of the supports for borrowers affected by the COVID-19. This time, financial institutions were urged to offer a set of minimum measures on debt restructuring options for borrowers according to their loan types. The options could include, for example, minimum repayment reduction, loan maturity extension, or reduction in loan installments. Furthermore, financial institutions were asked to not demand repayment of the entire loan principal and all interests suspended or installments reduced during the debt holiday at once when the measures expire. Also, late payment fees or any other charges will not be applied on borrowers participating in this program.

(3) Additional lending In order to support businesses with liquidity needs and those requiring changes to the business operations to prepare for the new economic environment, the BOT, authorized by the Financial Assistance Emergency Decree, would provide soft loans of 500 billion baht for financial institutions, who would then on-lend the funds to SMEs**. To be eligible for the loans, SMEs must have a normal repayment status and a credit line of not exceeding 500 million baht with each financial institution as of end-2019**. The maximum drawdown for the soft loan is 20% of the loan outstanding as of end-2019, and the interest rates on these loans must not exceed 2 percent[13/].

The soft loans would be distributed to borrowers across a wide range of sectors[14/], most notably SMEs (1) in the provinces, (2) in sectors impacted by COVID-19 such as trading, manufacturing, and service, (3) of small size with 20 million baht or less in existing credit lines, and (4) rated as medium to high-risks by financial institutions.

However, the amount of soft loans granted to date remained small compared to the size of the program, which is 500 billion baht. This was partly because financial institutions had already issued some new loans previously using supports from other government programs and continued to do so (Chart 1). These programs included the Government Savings Bank Soft Loan Scheme[15/], which comprised the financial institutions on-lending program and the direct SME lending program, and the credit guarantee facilities by the Thai Credit Guarantee Corporation[16/]. When new loans granted under the BOT soft loan program and other government SME loan support programs are combined, the amount of new loans would total over 200 billion baht. Moreover, as the economy began to recover and more businesses started to resume operations, the utilization of the BOT soft loans was expected to increase, supporting business liquidity needs and recovery.

[13/] BOT Notification No. BOT.FPG1.2/2020 Financial Assistance to Small and Medium-sized Enterprises Affected by the COVID-19 Pandemic (April 22, 2020) (Thai version only)

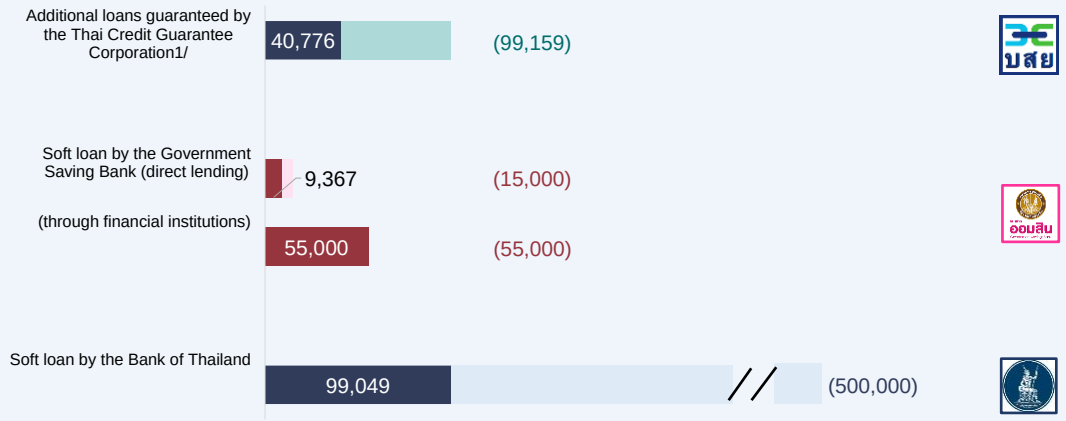
[14/] The Bank of Thailand's mission to assist individuals and businesses during the COVID-19 pandemic (May 28, 2020). (Thai version only)

[15/] Phase I Package of direct and indirect economic relief measures in response to impacts from Coronavirus (March 10, 2020) (Thai version only)

[16/] The Funding Thai SMEs for Thailand program (January 7, 2020) (Thai version only)

Chart 1 Progress on additional liquidity supports to SMEs

New loan in million baht, ( ) total loan amount of each measure



Note: 1/ Credit guarantee program implemented by the Thai Credit Guarantee Corporation since March 1, 2020 included
(1) Phase 8 of the Portfolio Guarantee Scheme : PGS8 and
(2) Phase 3 of the Credit Guarantee for Micro Entrepreneur

Source: Bank of Thailand (Data as of June 24, 2020)

With regard to retail borrowers, both existing and new, specialized financial institutions have implemented further credit support measures to assist their liquidity needs[17/]. For example, the Government Savings Bank has launched a targeted loan program for freelancers and salary workers, while the Bank for Agriculture and Agricultural Cooperatives has announced a 20 billion baht emergency loan program for farm households.

(4) Debt restructuring: The BOT announced the relaxation of supervisory regulations[18/] to encourage debt restructuring for business and household borrowers who were not yet classified as NPL. Restructuring options may include extension of loan repayment period, provision of additional working capital, interest reduction, or extension of loan maturity with lower interest rates that match declining income profile post-pandemic. Furthermore, the BOT relaxed rules in several areas. First, the classification of borrowers where the borrowers who were not yet classified as NPL can be immediately classified as normal if they could make repayments in accordance with the debt restructuring agreement. This restructuring would be considered pre-emptive, instead of Troubled Debt Restructuring (TDR). Moreover, the relaxation also applied to the classification of NPL borrowers who could comply with the restructuring agreement. Second, the loan loss provision rules were eased to facilitate debt restructuring for NPL borrowers.

The timeline of the debt restructuring program, as well as the debt holiday and reduction measures, are shown in Chart 2.

17/ Credit support measures for borrowers affected by the economic conditions and the COVID-19 pandemic, and the extension of new or additional credit lines.

18/ BOT Circular No. BOT.FPG (01)c 276/2020

Chart 2 The BOT issued a series of credit assistance and debt restructuring measures

Announcement date	Type of borrowers receiving assistance	Credit measures	Duration
Feb 28, 2020	All types of retail borrowers and SMEs	(1) Financial institutions implemented credit assistance measures including both **liquidity provision** and **debt restructuring** (2) **Relaxed regulations regarding classification of borrowers and levels of loan loss provision** so that financial institutions could accelerate the implementation of pro-active debt restructuring for borrowers	Jan 1, 2020 – Dec 31, 2021
Mar 26, 2020	All types of retail borrowers and small SMEs	Coordinate with financial institutions to issue **minimum debt moratorium and debt reduction requirements** for each type of borrowers	Depend on loan type
Apr 22, 2020	SMEs with a credit line not exceeding 100 million baht as of end-2019	**Extended** debt payment holiday	Apr 23 – Oct 22, 2020
Jun 19, 2020	All types of retail borrowers and small SMEs	Coordinate with financial institutions to introduce the **second phase of minimum measures to postpone and reduce debt repayment burdens** for each type of borrowers.	Depend on loan type

Looking ahead, the MPC emphasized the need to urge financial institutions to accelerate lending, using supports from the various measures implemented, in order to address liquidity problems in a targeted and timely manner. Moreover, while the government's financial and fiscal relief measures remain in place to shore up borrowers' liquidity, financial institutions should urgently implement a comprehensive debt restructuring program, particularly for household and SMEs, to make sure that they can make repayments in accordance with the debt restructuring agreements. Successful debt restructuring will contribute significantly to economic recovery going forward.

3. Monetary Policy Decision

The MPC cut the policy rate to 0.5 percent, the historically low level, in order to provide an appropriate and timely support from monetary policy accommodation

At the meeting on May 20, 2020, the Committee voted 4 to 3 to cut the policy rate by 0.25 percentage point from 0.75 to 0.50 percent. Three members voted to maintain the policy rate at 0.75 percent at this meeting. The Committee viewed that the Thai economy would experience a larger contraction in 2020 than previously assessed due to a more-severe-than-expected impact of the COVID-19 outbreak and strict containment measures implemented by governments in many countries including Thailand, which led to disruptions of global economic activities. Moreover, the impacts of the economic contraction on employment and financial system vulnerability would be more pronounced, as more people might be affected than expected, especially businesses that were unable to adapt at present or in response to changing business structure in the future. This would result in a slow economic recovery and a lower long-term potential economic growth. With regard to inflation, headline inflation would be more negative than previously assessed. Most members thus viewed that additional monetary policy accommodation was necessary to provide an effective and timely support to the real economy, as well as reinforce the previously implemented fiscal, financial, and credit measures. However, three members voted to maintain the policy rate at 0.75 percent, focusing on expediting the effectiveness of the announced financial and credit measures. Some members deemed it appropriate to wait to assess the effectiveness of the announced measures, as well as the outlook of the economy and the outbreak following the relaxation of the containment measures before implementing additional financial measures in the period ahead.

At the meeting on June 24, 2020, the Committee voted unanimously to maintain the policy rate at 0.50 percent. The Committee assessed that the Thai economy would contract in 2020 more than the previous assessment but would gradually recover in the second half of 2020 in line with improving economic activities in many countries, including Thailand, that could effectively contain the outbreak. Meanwhile, headline inflation would be more negative than previously assessed but would return to the target in 2021. The Committee viewed that the extra accommodative monetary policy since the beginning of the year as well as the fiscal, financial, and credit measures additionally announced helped alleviate adverse impacts and would support the economic recovery after the pandemic subsided. These would facilitate the return of inflation to the target, and reduce financial stability risks.

In determining the appropriate course of monetary policy, details on the key assessment are as follows.

Financial system stability became more vulnerable in line with the economic conditions

Financial system stability became more vulnerable in line with the economic conditions. The Committee deemed it necessary to monitor the followings. First, asset prices in global and Thai financial markets might experience sharp corrections as a result of changes in investors' perspectives toward the global economic outlook, especially those regarding the COVID-19 outbreak and trade tensions between the U.S. and China. In addition, corporate bonds might be downgraded to non-investment grade, which might result in forced selling

especially among investors that were subject to the policy on minimum risk tolerance. Second, debt servicing ability of businesses and households could deteriorate as global and Thai economies could contract more than expected, and some government relief measures would soon phase out, particularly remedy and living expense relief measures. This might result in higher non-performing loans and affect financial institutions stability. In this regard, the Bank of Thailand had announced the second phase of measures to support retail borrowers affected by the COVID-19. This would readily reduce risks of defaults.

Looking ahead, additional measures should be prepared for the highly uncertain COVID-19 outbreak and rising default risks of businesses and households

The fiscal, financial and credit measures recently announced helped ease the impacts of the COVID-19 outbreak on the private sector. However, financial system stability became more vulnerable given highly uncertain economic outlook due to the COVID-19 situation. Thus, **the implementation of financial and credit measures in the period ahead should place importance on the followings. First, liquidity in the financial system, which remained ample at the macroeconomic level, should be efficiently distributed to businesses and households affected by the pandemic,** by expediting credit extensions through several measures previously announced. **Second, debt restructuring for borrowers,** especially households and SMEs, **should be accelerated in order to create widespread impacts** while debt holidays and debt reduction measures were still in place. **Third, pre-emptive measures were necessary to cope with solvency risks of households and businesses.** This should take into account the differences in recovery potentials and competitiveness among various groups of households and businesses after the outbreak subsided. The Committee deemed it important to establish mechanisms to expedite multi-creditor debt restructuring, as well as expand roles of asset management companies (AMCs) in warehousing NPLs in the financial system to be appropriately managed. **Fourth, maintaining sound financial institutions stability would be necessary** to cope with uncertainties in the future and ensure the credit extension ability to restore the Thai economy after the COVID-19 outbreak situation subsided. Nevertheless, the BOT had requested for cooperation from commercial banks to suspend interim dividend payments in 2020 and stock buybacks, and any early redemption or purchases of debt instruments qualified as tier-1 or tier-2 capital prior to their maturities. This would ensure that commercial banks continue to maintain sufficient capital until the new capital fund management plan would be revised in response to the impacts of the outbreak and changing patterns of business operations after the outbreak subsided.

Targeted labor policies would help lessen the impacts of the COVID-19 outbreak on the long-term potential economic growth

The Committee expressed concerns over the employment outlook which could significantly decline and would take a long period of time before returning to normal, both for workers and the self-employed. Some workers were expected to be temporarily unemployed particularly during when containment measures were still in place. Meanwhile, some workers would be more affected and face difficulties in returning to the labor market even after the outbreak subsided, especially those in recently-graduated, old-aged, and unskilled groups. This was because businesses had high level of excess capacity and thus adjusted by increasingly adopting digital technology and automation in the production process. This would result in economic scars on the long-term potential economic growth.

In recent periods, the government had continuously implemented measures to remedy affected workers. Looking ahead, **as the government relaxed containment measures and started to boost domestic demand and tourism in order to restore economic activities, labor policies should adapt their roles from remedying affected workers to creating jobs and accelerating upskill of workers, as well as wage subsidies for businesses to maintain employment.** Nevertheless, design of labor policies in the period ahead should not only focus on widespread support, but should also be targeted. This would partly help reduce the economic scars caused by the COVID-19, and support a strong and sustainable economic restructuring. **(Box:** *Restoring Thailand's labor markets from impacts of the COVID-19)*

There was still fiscal policy space to facilitate restoration and restructuring of the economy such that the government should place importance on supply-side policies to support economic restructuring to be able to recover in the new economic landscape after the COVID-19 pandemic subsided

The Committee assessed that there remained fiscal policy space to support economic restoration and restructuring in the period ahead. This was reflected in Thailand's low level of public debt prior to the COVID-19 outbreak with average debt maturities of 10 years, most of which were denominated in Thai baht. Nevertheless, fiscal measures and public investment must be major tools in shoring up the economy, as well as reducing risks of substantial economic contraction to the level that would pose risks to economic and financial system stability. Such implementation of measures could increase the level of public debt-to-GDP ratio, despite at a manageable level. Supporting factors would come from considerably low government's funding costs in line with lower government bond yields following a series of policy rate cuts.

Nevertheless, **government should focus on policies to upgrade the potential growth for long-term benefits. Particularly, government should accelerate the implementation of supply-side policies to support economic restructuring and changing patterns of business operations to the new economic landscape.** This was because businesses in each sector would face varying manners of recovery and challenges for the adjustments after the outbreak subsided. This might lead some industries to be unable to fully recover and have excess capacity. The government would need to (1) establish mechanisms to manage excess capacity in industries such as policies to facilitate consolidation among businesses with substantial excess capacity especially in tourism-related industries or policies to promote adjustments of business models to the post-COVID environment, (2) design policies to support and enhance the competitiveness of high-potential targeted industries with fast recovery potential after the pandemic subsided such as providing incentives for the private sector to invest through the fiscal, financial, and credit measures, as well as relaxing government regulations regarding business operations, (3) expedite infrastructure investment especially public-private partnership projects both within and outside the Eastern Economic Corridor (EEC), together with infrastructure for the digital economy, (4) facilitate reskilling and upskilling of labor to support employment in targeted industries and adjustments to the post-COVID environment, and (5) accelerate reform of government regulations that are outdated or obstructive to the adjustments of businesses to the new economic landscape. *(***Box:** *Roles of public policies in restoring the Thai economy amid the highly uncertain COVID-19 outbreak)*

The Committee would stand ready to use additional appropriate policy tools if necessary.

Looking ahead, the Committee would monitor developments of economic growth, inflation, and financial stability, together with associated risks, including external risks, the impacts of the COVID-19 outbreak, and the effectiveness of fiscal, financial, and credit measures, as well as the baht situation which might affect the economic recovery in deliberating monetary policy going forward. The Committee would stand ready to use additional appropriate policy tools if necessary.

Roles of public policies in restoring the Thai economy amid the highly uncertain COVID-19 outbreak

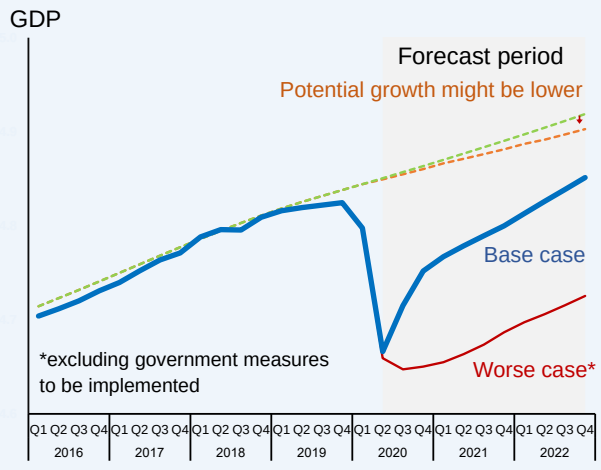
The COVID-19 outbreak led to unprecedentedly severe impact on the economy and the way of life. Consequently, governments worldwide had to impose strict containment measures to control the spread of the virus, causing temporary disruptions to economic activities across several sectors. It was the first time in the history that almost all economic engines shut down simultaneously. It could thus take several years before economic activities return to the pre-pandemic level.

However, the Thai economy, compared to others, still have strong buffers in various aspects. **(1) External stability remained solid**, with high levels of current account surplus and international reserves, as well as a low level of external debt to GDP. **(2) Fiscal space remained adequate** for the government to provide relief measures and support economic recovery, with public debt outstanding at 43 percent of GDP in April 2020, well below the international benchmark of 60 percent. **(3) Financial institutions stability remained robust**, as reflected in high levels of commercial banks' capital funds and reserves.

Despite the presence of these strong buffers, vulnerabilities existed in certain sectors. In particular, household debt remained high. SMEs also faced intensifying competition from large businesses, e-commerce and rapid technological changes. These existing vulnerabilities were aggravated by the COVID-19 situation. Hence, the use of abovementioned buffers would be crucial to shore up the economy in the period ahead. **Policymakers would need to take into account current key challenges, as well as should be forward-looking to changing environment after the COVID-19 outbreak subsides.** Roles of public policies should address the following aspects:

1. Role in providing short-term economic support

Chart 1 Projections of the Thai economic recovery path



The public sector has employed buffers of the fiscal and financial institutions to urgently provide liquidity support and ease financial burdens of households and businesses, especially in the initial phase when a la rge number of people were affected by the COVID-19 containment measures. The government introduced large-scale relief measures to alleviate and remedy those affected by the outbreak in the short run. Meanwhile, the Bank of Thailand cut the policy rate to th e historically low level and introduced credit measures to assist borrowers through financial institutions, such as debt repayment moratorium and reduction, debt restructuring, and soft loans. Such policy synchronization would help support the Thai economic recovery in the medium term, in line with the baseline projection (the blue line in Chart 1).

As the baseline projection suggested, **despite the massive government stimulus, it would take some time before the Thai economy could return to the pre-COVID-19 level.** This was because the income shock has a long-term impact on the private sector's

financial positions. In addition, businesses, especially those in the tourism sector, have to alter business models in response to the new normal where people became more concerned on health safety, at least until vaccines and medicines become readily available. **Hence, in the next period in which the outbreak would be contained and the government starts to relax containment measures, public policies would have a vital role in supporting the economic recovery toward the pre-pandemic level.** Fiscal and supply-side policies, together with financial and credit measures, would be of particular importance.

2. Role in mitigating risk of the worst case scenario and maintaining the long-term potential of the Thai economy.

The worst case scenario should be developed in order to prepare for risks that could arise amid the highly uncertain global environment, as well as assess the adequacy of buffers, especially **if the economy contracts significantly this year and is unable to recover to its previous growth path in the medium term** (the red line in Chart 1). This could happen as a result of (1) external factors, such as the global spread of COVID-19 which could cause a more severe and prolonged global economic contraction, and trade tensions between the U.S. and China that could intensify and further weigh on Thai exports and tourism, and (2) domestic factors, such as the second wave of the COVID-19 outbreak that could prompt the government to re-introduce strict containment measures and prevent the introduction of travel bubbles. This worst case scenario did not include additional government measures to be implemented. Thus, economic developments and the COVID-19 situation should be closely monitored through real-time economic indicators.

Should the worst case scenario materialize, the damage would not be limited only to the short-term impact, but would also leave the long-term economic scars even after the outbreak subsides. This would, in turn, cause Thailand's potential economic growth to decline (the orange dotted line in Chart 1). The long-term impact could stem from risks in both the economy and the financial system. Examples include, first, some businesses may become bankrupt and permanently closed, while businesses that survive may decide to reduce or delay their investments. Second, many businesses may reduce employment, causing workers to face greater difficulties in finding new jobs, especially those who lack the skills required by other industries. Third, there could be massive defaults of corporates and households, resulting in much higher levels of non-performing loans (NPL) and rendering inadequate commercial banks' capital and loan loss provision to cover risks from new loans in the period ahead. This could cause a macro-financial negative feedback loop on the economy in the long term.

Although the existing buffers remained strong, the public sector should optimize the use of these buffers to ensure the fullest efficiency and should be prepared for any case of emergency. An emphasis should be given to factors that could have an extensive impact on financial institutions, businesses and households, and to variables with large multiplier effects or high linkage with various sectors of the economy. **To mitigate the risk of the worst case scenario and enhance Thailand's potential economic growth, it is imperative to focus on the following issues:**

1. Preventing the spillover of economic problems to the financial system, both financial markets and financial institutions system. The Bank of Thailand accelerated certain actions which include the followings. First, a timely implementation of measures was done to stabilize the financial markets in order to shore up investor confidence and reduce financial market volatility, such as purchases of government bonds to maintain liquidity in the

bond market, the establishment of the Mutual Fund Liquidity Facility (MFLF) and the Corporate Bond Stabilization Fund (BSF). Second, financial institutions were encouraged to expedite debt restructuring to ensure that borrowers would be able to repay their debt under the revised conditions, which would reduce risk of NPL and help support the stability of financial institutions system. Third, the Bank of Thailand requested cooperation from commercial banks to suspend interim dividend payments in 2020, stock buybacks, and any early redemption or purchases of debt instruments qualified as tier-1 or tier-2 capital prior to their maturities to ensure that commercial banks sustain strong capital. Such implementation would help shore up confidence in the financial markets and financial institutions system, as well as ensure sound preparation for coping with the risks in the future.

2. Designing supply-side policies to support the economic restructuring both policies related to business operation, capital reallocation, and labor to be consistent with the post-COVID environment. Once the COVID-19 outbreak subsides, businesses in different sectors would face varying manners of recovery and difficulties regarding adjustment. This might lead some industries to be unable to fully recover and have excess capacity. The government should, first, establish mechanism to manage excess capacity in the Thai manufacturing sector, such as policies to facilitate mergers and acquisitions among businesses with substantial excess capacity, particularly those in tourism-related businesses, or policies to promote adjustments of business models to suit the post-COVID environment. Second, the design of policies to support and enhance the competitiveness of high-potential targeted industries with fast recovery potential after the pandemic subsides could be done through, for example, providing incentives through the use of fiscal, financial, and credit measures for the private sector to invest, as well as relaxing government regulations that remain obstructive to business operations. Third, the government should also expedite infrastructure investment especially public-private partnership projects both within and outside the Eastern Economic Corridor (EEC), together with infrastructure for the digital economy. Fourth, reskilling and upskilling of workers should be promoted to support employment in targeted industries and adjustments of businesses to suit the post-COVID environment. Fifth, reforms of government regulations that are outdated or obstructive to the adjustments of businesses to the new economic landscape should be accelerated.

In conclusion, although Thailand is currently able to contain the spread of COVID-19 and has adequate buffers to implement financial and fiscal measures to address the impacts of the COVID-19, it is however necessary to stand ready to introduce measures to cope with risks in the period ahead in a more targeted, timely and effective manner. This would ensure adequate buffers and reduce risk of economic scars should the worst case scenario materialize. In addition, this would help sustain the long-term potential economic growth. The implementation of these measures would require policy synchronization between public health policies, the fiscal, financial and credit measures, as well as supply-side policies, in order to support economic restructuring to be consistent with the COVID-19 situation in a given period.

Restoring Thailand's labor markets from impacts of the COVID-19

The COVID-19 situation significantly affected economic activities and Thai labor markets. During the beginning of the outbreak, some businesses had to temporarily suspend business operations, which resulted in a widespread impact on labor. At present, the government was successful in containing the spread and thus, businesses could gradually resume operations. However, given the highly uncertain situation as well as changing business models under the new normal, this could add pressures on the recovery of domestic demand and labor markets. **Measures in the period ahead should support the adjustments of labor to foster long-term sustainable economic growth.**

During the beginning of the outbreak, most labor policies were short-term measures to remedy affected workers. These measures can be divided into 2 groups according to the status in the Social Security System. (Chart 1) **(1) Workers in the Social Security System have been remedied according to their rights** such as rights to receive compensation from the Social Security Fund in the event of unemployment,[19/] job termination due to force majeure,[20/] or temporary suspension of business operation according to section 75 that is not due to force majeure.[21/] According to the latest data in June 2020, the Social Security Fund has remedied approximately 2 million affected workers. **(2) Workers outside the Social Security System have been assisted by the government through formal registration or application platform,** such as 15 million freelancers under the "Rao Mai Ting Gun" (You will never be left behind) project, as well as 8 million farmers, that were entitled to receive benefits of 5,000 baht for 3 months, and employment of newly graduates as demanded by government agencies.

Chart 1 Supports for labors affected by the COVID-19, classified by group of labors

Group of labors	Impact	Assistance from businesses	Public assistance
Labors in the social security system:			
1. Unemployed person	Unemployment and income loss	-	Employees are entitiled to social security benefits for an additional period of 6 months
2. Work cessation due to force majeure	Job positions maintained, but cease working and suffer income loss	Businesses may ask employees to leave without pay	The Social Security Office compensates at a rate of 62% of daily wages for a period of 3 months
3. Temporary work cessation under section 75	Job positions maintained, but cease working and suffer *partial* income loss	Employers have to compensate at least 75% of daily wages during the work cessation period	-
4. Adjustment to hourly employment contract (old and new employees)	Job positions maintained, but work for *fewer* hours and suffer *partial* income loss	Employees get paid on an hourly basis.	-
Labors outside the social security system:			
5. Freelancers/ workers outside the social security system	Work for *fewer* hours and suffer *partial* income loss	-	"Rao Mai Ting Gun" project (The government offers 5,000-baht cash handouts per person for 3 months.)
6. Farmers	Work for *fewer* hours and suffer *partial* income loss	-	Relief measure for farmers (The government offers 5,000-baht cash handouts for 3 months.)
7. Newly graduates	Unable to find job and suffer income loss	-	The government considers hiring new graduates as requested by government agencies.

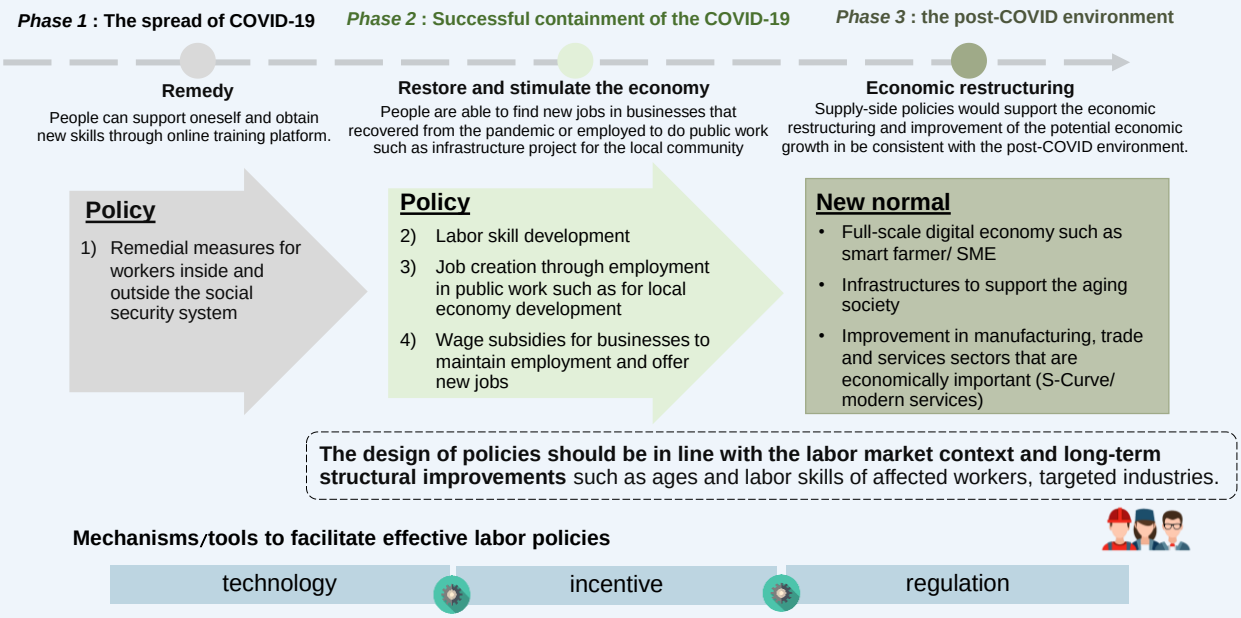
[19/] Section 38of the Social Security Act B.E. 2533

[20/] Ministerial Regulations for receiving unemployment benefits due to force majeure caused by the pandemic of dangerous communicable diseases under the communicable disease law B.E. 2563)April, 17,2 020(

[21/] Section 75 of the Labor Protection Act B.E. 2541

As for the second phase that the government began to successfully contain the spread (Chart 2), the government has relaxed the containment measures and implemented measures to boost consumption and domestic tourism to restore economic activities and facilitate employment of businesses. However, workers could be severely affected and find it difficult to return to the labor markets. These include workers in businesses with excess production capacity due to severe impacts and in businesses that adopted automation in the production process, as well as the newly graduated, aging, and low-skilled workers. **Therefore, labor policies should adapt their roles from remedying to employing and enhancing skills of labor.** (Chart 2) These policies include **(1) job creation policy** to support unemployed and newly-graduated workers, especially employment at the community level due to public work for local economy development, which would partly help create job for non-farm workers to return home, **(2) labor skill development policy** along with subsidies to motivate workers to reskill and upskill through various learning channels including online training platforms,[22/] which can be continuously implemented in the long run so that workers will be familiar with lifelong learning and immuned to survive in the era of high uncertainties, **(3) wage subsidies for businesses to support employment** in the case that businesses does not cut employment or cut to the level not exceeding the specified threshold. This would ensure that workers remain employed and businesses could still operate. In recent periods, the government has already implemented some of these policies, where the Social Security Fund would compensate workers on behalf of the employers in case of job termination due to force majeure. Nevertheless, wage subsidies should be implemented together with providing incentives for the private sector to adjust to the new normal, including the move toward the digital economy, the aging society, and modern services.

Chart 2 Labor policies after the COVID-19 outbreak subsides should focus on stimulating the economy, creating jobs, and promoting an upskill of labor to support changing economic structure in response to the new economic landscape



[22/] Kampon Phornphattanapaisarnkul, 2020, "Labor Market Digital Transformation: Anti-crisis path" FOCUSED AND QUICK (FAQ) Issue171 , Bank of Thailand

Nevertheless, the design of labor policies in the period ahead should not only focus on widespread support, but should also be appropriately and clearly targeted, such as ages and skills of labors, as well as targeted industries through wage subsidies for businesses to support employment. The support should aim at small- and medium-sized businesses in S-curve and New S-curve industries that would employ young-aged workers or the first-jobbers so that this group of workers will be allowed to enhance skills, especially digital skills.

More importantly, policy makers should push forward broad-based measures that will be most effective through 3 key development mechanisms. These include **(1) technology development,** where cooperation between the government and private sector is necessary including job-search platform and upskilling of labor to match with job vacancies, as well as suggestions on labor skills to be further enhanced, **(2) provision of incentives for workers and employers to participate in skill development policy as needed or according to the national strategic plan** including allowing employers to link data on job vacancies, qualifications, and labor skills needed with the platform so that the system could correctly match unemployed workers with job vacancies, **(3) adjustment of some regulations to suit the new normal** including hourly employment or more flexible employment.

In conclusion, the government has implemented measures to remedy both workers within and outside the Social Security System affected during the beginning of the outbreak to a certain level. In the period ahead, labor policies would be more important in restoring and boosting the economy through job creation policy, skill development policy, and wage subsidies for businesses to support employment. This would partly help mitigate the economic scars from the impact of COVID-19 and support the strong and sustainable economic restructuring.

4. Appendix

4.1 Table

Thai Economy Dashboard

Percent	2018	2019	2018	2019				2020
			Q4	Q1	Q2	Q3	Q4	Q1
GDP growth	4.2	2.4	3.8	2.9	2.4	2.6	1.6	-1.8
Production								
Agriculture	5.5	0.1	1.3	1.7	-1.4	2.7	-1.6	-5.7
Non-agriculture	4.0	2.6	4.1	3.0	2.8	2.5	2.0	-1.4
Manufacturing	3.2	-0.7	3.8	0.2	0.1	-0.8	-2.3	-2.7
Construction	2.4	2.0	3.0	3.0	3.4	2.7	-1.9	-9.9
Wholesales and retail trade	6.6	5.7	7.3	6.7	5.5	5.3	5.2	4.5
Accommodation and Food Service	7.6	5.5	5.1	5.0	3.6	6.7	6.8	-24.1
Transport and storage	4.4	3.4	4.0	3.7	3.0	3.1	3.9	-6.0
Information and Communication	7.6	8.8	7.3	7.1	8.8	8.2	10.8	4.4
Financial intermediation	3.4	2.7	1.9	2.0	1.8	3.8	3.4	4.5
Real estate and renting	5.7	3.4	5.4	5.0	3.3	2.2	3.0	1.6
Expenditure								
Domestic demand	4.0	3.3	4.3	4.1	3.4	3.4	2.3	-0.5
Private consumption	4.6	4.5	5.1	4.8	4.7	4.3	4.1	3.0
Private investment	4.1	2.8	5.7	4.3	2.1	2.3	2.6	-5.5
Government consumption	2.6	1.4	2.2	3.5	1.5	1.7	-0.9	-2.7
Public investment	2.9	0.2	-0.3	0.0	1.5	3.7	-5.1	-9.3
Imports of goods and services	8.3	-4.4	4.6	0.1	-3.4	-5.9	-8.3	-2.5
imports of goods	7.9	-5.5	3.8	-2.8	-3.7	-6.8	-8.6	-0.3
imports of services	9.9	0.0	7.7	13.0	-2.1	-2.1	-7.1	-10.5
Exports of goods and services	3.3	-2.6	1.3	-3.5	-4.0	0.6	-3.6	-6.7
exports of goods	3.8	-3.6	1.1	-4.4	-4.6	-0.1	-5.1	2.0
exports of services	2.0	0.4	2.0	-1.1	-1.7	3.2	1.1	-29.8
Trade balance (billion, U.S. dollars)	22.4	26.7	4.6	6.7	6.0	8.1	6.0	8.0
Current account (billion, U.S. dollars)	28.5	38.4	6.3	12.4	5.4	9.8	10.7	8.9
Financial account (billion, U.S. dollars)	-14.9	-15.0	-2.5	-5.1	-2.5	-1.1	-6.2	-6.9
International reserves (billion, U.S. dollars)	205.6	224.3	205.6	212.2	215.8	220.5	224.3	226.5
Unemployment rate (%)	1.1		1.1	0.9	1.0	1.0	1.0	1.0
Unemployment rate, seasonally-adjusted (%)	n.a.		1.3	0.9	0.9	1.0	1.0	1.0

Source: Office of the National Economic and Social Development Board National Statistical Office and Bank of Thailand

Financial Stability Dashboard

Indicators	2018	2019	2018	2019				2020		
			Q4	Q1	Q2	Q3	Q4	Q1	Apr	May
1. Financial market sector										
Bond market										
Bond spread (10 years - 2 years)	0.7		0.7	0.7	0.4	0.1	0.3	0.6	0.5	0.7
Equity market										
SET index (end of period)	1,563.9	1,563.9	1,563.9	1,638.7	1,730.3	1,637.2	1,579.8	1,125.9	1,301.7	1,342.9
Actual volatility of SET index[1/]	12.1	14.0	14.0	8.4	7.8	10.7	10.2	47.0	32.3	17.5
Price to Earnings ratio (P/E ratio) (times)	14.8	14.8	14.8	17.1	18.6	18.7	19.4	14.3	14.6	13.3
Exchange rate market										
Actual volatility of Thai baht (%annualized)[2/]	4.6	4.1	4.8	4.6	4.5	4.6	2.7	5.5	6.6	4.7
Nominal Effective Exchange Rate (NEER)	115.6	123.2	117.0	119.9	120.9	125.0	126.8	122.4	120.9	122.6
Real Effective Exchange Rate (REER)	107.2	112.8	108.3	110.1	111.3	114.6	115.3	107.7	106.7	107.9
2. Financial institution sector[3/]										
Minimum Lending Rate (MLR)[4/]	6.28		6.28	6.28	6.28	6.28	6.08	5.93	5.53	5.36
12-month fixed deposit rate[4/]	1.37		1.37	1.42	1.46	1.45	1.33	0.75	0.75	0.54
Capital adequacy										
Capital funds / Risk-weighted asset (%)	18.3	19.6	18.3	18.2	18.0	19.2	19.3	18.4	18.9	n.a.
Earning and profitability										
Net profit (billion, Thai baht)	207.2	270.9	49.5	57.1	60.8	96.5	56.4	52.9	n.a.	n.a.
Return on assets (ROA) (times)	1.1	1.4	1.1	1.2	1.2	2.0	1.4	n.a.	n.a.	n.a.
Liquidity										
Loan to Deposit and B/E (%)	98.2	95.5	98.2	96.8	97.9	97.1	95.5	92.5	92.7	n.a.
3. Household sector										
Household debt to GDP (%)	78.6	79.5	78.4	78.1	78.1	78.5	79.9	80.1	n.a.	n.a.
Financial assets to debt (times)	n.a.	n.a.	2.5	n.a.	n.a.	n.a.	2.9	n.a.	n.a.	n.a.
Non-Performing Loans (NPLs) of commercial banks (%)										
Consumer loans	2.7	2.9	2.7	2.8	2.9	2.8	2.9	3.2	n.a.	n.a.
Housing loans	3.2	3.7	3.3	3.5	3.7	3.5	3.7	4.0	n.a.	n.a.
Auto leasing	1.7	1.9	1.7	1.9	1.9	1.9	1.9	2.1	n.a.	n.a.
Credit cards	2.3	2.4	2.3	2.7	2.4	2.4	2.4	3.5	n.a.	n.a.
Other personal loans	2.5	2.3	2.5	2.4	2.3	2.4	2.3	2.6	n.a.	n.a.
4. Non-financial corporate sector[5/]										
Operating profit margin (OPM) (%)	7.5	6.8	6.4	7.5	5.7	7.4	6.7	6.4	n.a.	n.a.
Debt to Equity ratio (D/E ratio) (times)	0.7	0.7	0.7	0.7	0.8	0.8	0.7	0.8	n.a.	n.a.
Interest coverage ratio (ICR) (times)	6.2	4.4	5.3	5.1	3.6	4.9	4.0	4.0	n.a.	n.a.
Current ratio (times)	1.6	1.6	1.6	1.6	1.6	1.6	1.6	1.6	n.a.	n.a.
Non-Performing Loans (NPLs) of commercial banks (%)										
Large businesses	1.7	1.6	1.7	1.6	1.7	1.7	1.6	1.5	n.a.	n.a.
SMEs	4.6	4.6	4.6	4.7	4.5	4.8	4.6	4.8	n.a.	n.a.

Note:

[1/] Calculated by 'annualized standard deviation of return' method

[2/] Daily volatility (using exponentially weighted moving average method)

[3/] Based on data of all commercial banks

[4/] Average value of 5 largest Thai commercial banks

[5/] Only listed companies on Stock Exchange of Thailand (median value); with data revisions

Financial Stability Dashboard (continue)

Indicators	2018	2019	2018	2019				2020		
			Q4	Q1	Q2	Q3	Q4	Q1	Apr	May
5. Real estate sector										
Number of approved mortgages from commercial banks (Bangkok and Vicinity) (units)										
Total	73,361	70,876	21,623	19,710	15,983	17,141	18,042	15216	4,187	n.a
Single-detached and semi-detached hou	15,912	16,167	4,610	4,446	3,460	4,136	4,125	3626	973	n.a
Townhouses and commercial buildings	25,042	24,763	7,495	7,290	5,719	6,129	5,625	5330	1,342	n.a
Condominiums	32,407	1,718	9,518	7,974	6,804	6,876	8,292	6260	1,872	n.a
Number of new housing units launched for sale (Bangkok and Vicinity) (units)										
Total	124,977	116,933	35,861	29,498	25,999	27,182	34,254	17,513	2,736	4,471
Single-detached and semi-detached hou	18,311	19,423	5,050	4,458	4,213	5,014	5,738	3,244	615	1,632
Townhouses and commercial buildings	32,349	31,423	10,385	5,236	7,940	9,322	8,925	7,751	1,720	2,454
Condominiums	74,317	66,087	20,426	19,804	13,846	12,846	19,591	6,518	401	385
Housing price index (2009 = 100)										
Single-detached houses (including land)	138.6	144.0	139.0	141.9	142.9	145.0	146.3	149.2	149.7	n.a
Townhouses (including land)	149.5	156.4	151.1	156.6	154.7	154.6	159.8	164.2	164.2	n.a
Condominiums	180.9	182.3	185.9	181.6	177.0	179.0	191.6	189.6	190.4	n.a
Land	173.6	172.8	177.6	175.5	168.1	167.1	180.6	182.0	181.8	n.a
6. Fiscal sector										
Public debt to GDP (%)	41.8	41.2	41.8	41.8	41.3	41.1	41.2	41.7	42.9	n.a
7. External sector										
Current account balance to GDP (%)[6/]	5.6	7.0	4.8	9.3	4.1	7.2	7.5	6.6	n.a	n.a
External debt to GDP (%)[7/]	35.4	34.1	35.4	35.0	34.7	34.1	34.1	32.4	n.a	n.a
External debt (billion, U.S. dollars)	162.4	171.4	162.4	164.5	166.7	167.2	171.4	165.5	166.6	168.7
Short-term (%)	39.0	35.0	39.0	38.5	37.5	35.4	35.0	36.0	36.2	36.3
Long-term (%)	61.0	65.0	61.0	61.5	62.5	64.6	65.0	64.0	63.8	63.7
International reserves / Short-term external d	3.2	3.7	3.2	3.4	3.5	3.7	3.7	3.8	3.9	3.9

Note:

[6/] Current account / Nominal GDP at the same quarter

[7/] External debt / 3-year average nominal GDP

Table: Probability distribution of GDP growth forecast

Percent	2020				2021				2022
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1
> 20	0	0	0	0	0	2	0	0	0
16-20	0	0	0	0	0	9	2	0	0
12-16	0	0	0	0	0	21	9	3	3
8-12	0	0	0	0	0	23	20	13	12
4-8	0	0	0	0	2	19	23	22	22
0-4	0	0	0	2	12	13	19	22	22
(-4)-0	100	0	3	12	23	7	13	18	18
(-8)-(-4)	0	1	17	25	23	3	8	11	12
(-12)-(-8)	0	15	32	25	18	1	4	6	6
(-16)-(-12)	0	40	27	18	12	0	1	3	3
(-20)-(-16)	0	30	14	11	6	0	0	1	1
(-24)-(-20)	0	11	5	5	3	0	0	0	0
(-28)-(-24)	0	2	1	2	1	0	0	0	0
< (-28)	0	0	0	1	0	0	0	0	0

Table: Probability distribution of headline inflation forecast

Percent	2020				2021				2022
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1
> 6.0	0	0	0	0	0	1	1	1	1
5.0-6.0	0	0	0	0	0	3	2	2	2
4.0-5.0	0	0	0	0	0	8	6	6	5
3.0-4.0	0	0	0	0	1	14	11	11	11
2.0-3.0	0	0	0	1	2	20	18	17	16
1.0-2.0	0	0	1	4	6	20	20	19	19
0.0-1.0	100	0	5	9	12	16	18	17	17
(-1.0)-0.0	0	1	14	18	19	10	13	13	13
(-2.0)-(-1.0)	0	10	24	23	21	5	7	8	8
(-3.0)-(-2.0)	0	33	26	22	18	2	3	4	5
(-4.0)-(-3.0)	0	38	19	14	12	1	1	2	2
(-5.0)-(-4.0)	0	15	9	7	6	0	0	1	1
(-6.0)-(-5.0)	0	2	3	2	2	0	0	0	0
< -6.0	0	0	1	1	1	0	0	0	0

Table: Probability distribution of core inflation forecast

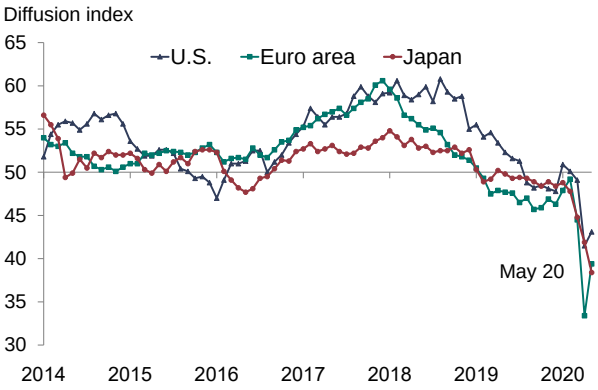
Percent	2020				2021				2022
	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1
> 2.5	0	0	0	0	0	0	0	0	0
2.0-2.5	0	0	0	0	0	1	1	1	1
1.5-2.0	0	0	0	1	2	3	4	3	3
1.0-1.5	0	0	1	4	5	8	10	8	8
0.5-1.0	100	10	8	11	10	14	15	14	13
0.0-0.5	0	40	24	20	16	18	17	16	15
(-0.5)-0.0	0	38	32	24	19	18	16	15	14
(-1.0)-(-0.5)	0	11	23	20	18	15	13	13	12
(-1.5)-(-1.0)	0	1	9	12	13	10	10	10	10
(-2.0)-(-1.5)	0	0	2	6	8	6	6	7	8
(-2.5)-(-2.0)	0	0	0	2	4	3	4	5	6
(-3.0)-(-2.5)	0	0	0	0	2	2	2	3	4
(-3.5)-(-3.0)	0	0	0	0	1	1	1	2	2
< -3.5	0	0	0	0	0	0	1	2	3

4.2 Chart Pack

Global Economy

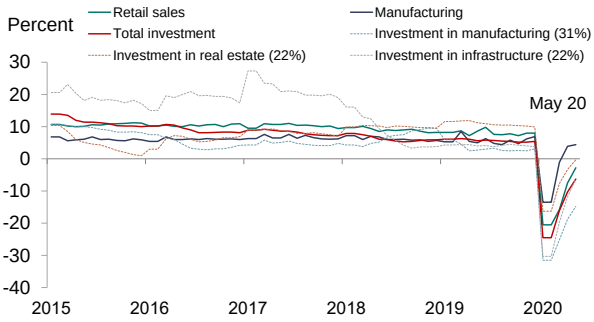
Governments in many countries worldwide implemented containment measures to control the spread of COVID-19, causing disruptions to global economic activities, especially in major advanced economies and Asian economies. Also, the Chinese economy grew at a slower pace. However, the series of financial and fiscal measures that had been implemented so far would help alleviate the economic impact and support recovery in the coming periods.

Manufacturing Purchasing Manager Index



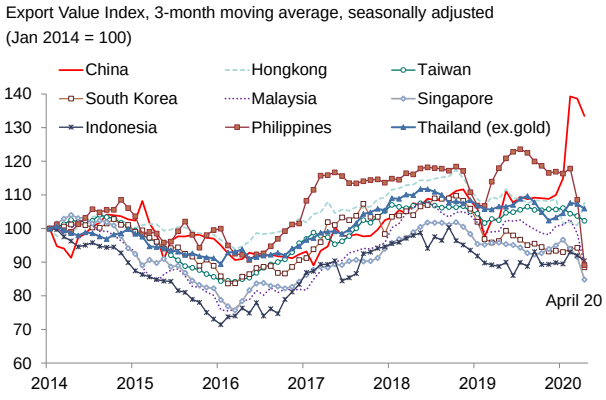
Sources: Bloomberg and Eurostat

China's economic indicators (Change from the same period last year)



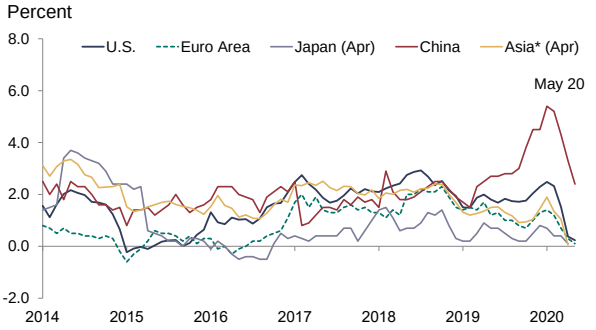
Note: ( ) denotes share to total investment
Source: CEIC

Asian exports



Source: CEIC

Headline inflation of Thailand's major trading partners

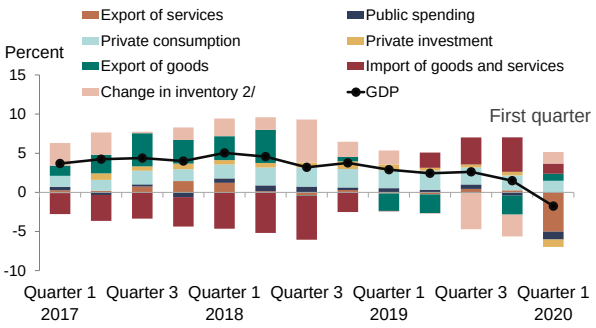


Note: *Average of headline inflation in Indonesia, South Korea, Malaysia, the Philippines, Singapore, and Taiwan
Source: CEIC

Thai economy

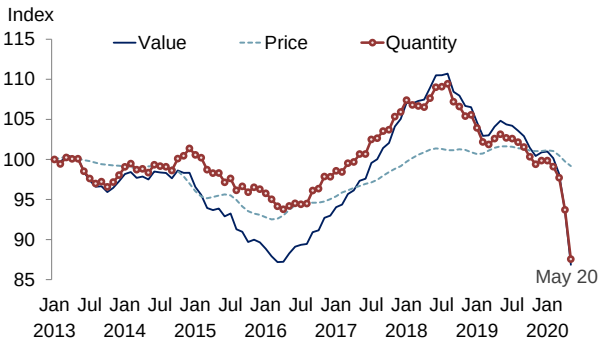
In light of the COVID-19 pandemic that became more severe and widespread, the Thai economy in the first quarter of 2020 contracted due to both international and domestic demand. Also, in the second quarter of 2020, the Thai economy was projected to contract sharply as a result of strict containment measures both in Thailand and abroad.

Contribution to Thailand's GDP growth[1/]



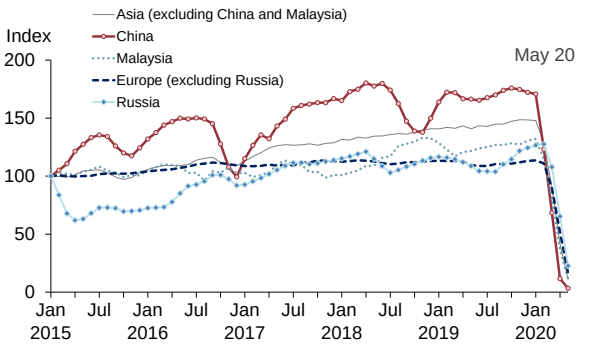
Note: [1/] Calculated by Chain Volume Measure method (CVM)
[2/] Change in inventory and statistical discrepancy
Source: Office of National Economic and Social Development Council, calculations by the Bank of Thailand

Thai exports (excluding gold): value, price and quantity
(3-month moving average, seasonally adjusted; January 2013 = 100)



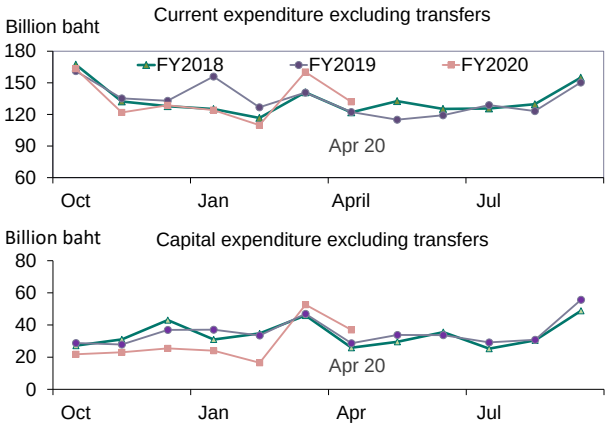
Source: Customs Department and Ministry of Commerce, calculations by Bank of Thailand

Index of foreign tourists classified by nationality
(three-month moving average, seasonally adjusted; January 2015 = 100)



Source: Department of Tourism

Public spending by central government

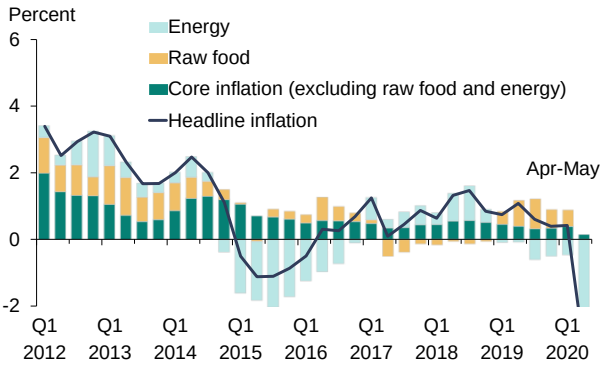


Source: Bureau of Budget, Fiscal Policy Office

Inflation

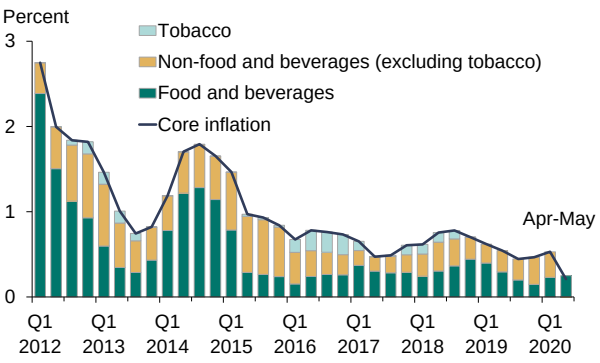
Headline inflation was negative in April and May mainly owing to the decline in energy prices. However, it was expected to be less negative in June on account of rising energy and fresh food prices. Meanwhile, core inflation softened due to the reduction in electricity and water bills, which was part of the government's measures to ease the burden on households. Regarding inflation expectation, short-term expectations edged up while long-term expectations declined slightly.

Contribution to headline inflation



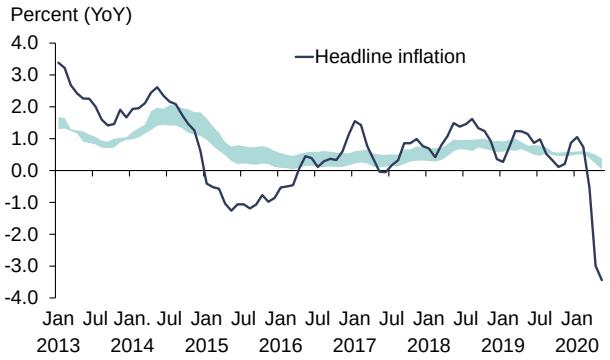
Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

Contribution to core inflation



Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

Underlying inflation indicators

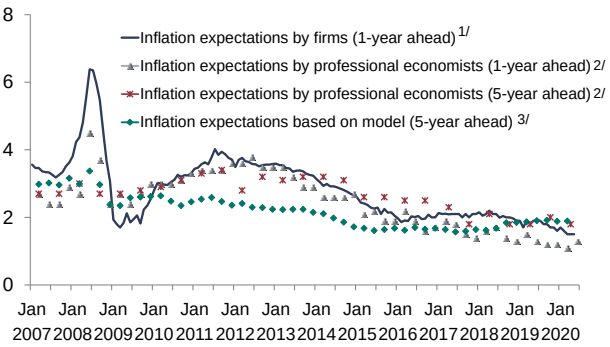


Note: The field shows the highest and lowest outcomes among different measures of underlying inflation. The measures included 1. **Trimmed mean** (excludes goods and services with most volatile price changes, removing the bottom 15 percentile and the top 10 percentile), 2. **Principal component model** (calculates changes in common statistical components that attribute price movements across categories of goods and services) and 3. **Core inflation excluding rents and government measures.**

Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

Inflation expectations

Percent change from the same period last year

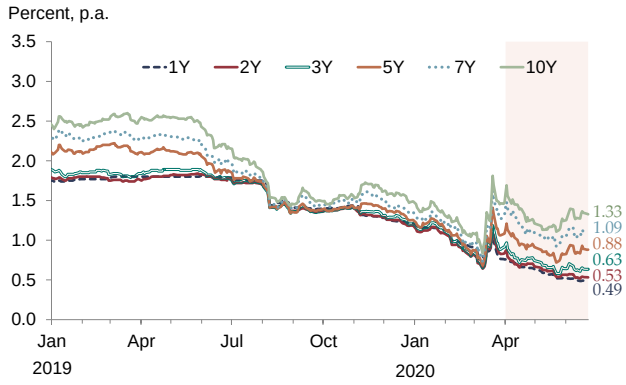


Source: 1/ Business Sentiment Survey (BSI) by the Bank of Thailand
2/ Asia Pacific Consensus Forecast
3/ Calculations based on macro-finance term structure model using bond yield and macroeconomic data

Financial conditions

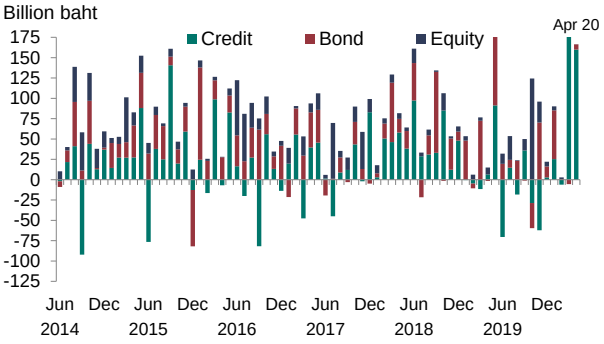
Short-term money market rates, lending rates, and deposit rates declined persistently, in line with the reduction in the policy rate and the Financial Institutions Development Fund (FIDF) contribution. Short-term government bond yields fell, but the government bond yield curve steepened as long-term yields fell by less than short-term yields. Meanwhile, corporate bond yields continued to increase. Private sector financing accelerated on account of the need for liquidity buffer among businesses affected by the COVID-19 pandemic, and also as a result of the government's soft loan programs for SMEs. However, household credit growth continued to soften. With regard to exchange rates, the baht against the U.S. dollar and the effective exchange rate appreciated throughout the second quarter, owing to the relaxation of containment measures in the region and sales of the U.S. dollars by domestic players.

Government bond yields



Source: Thai Bond Market Association (Thai BMA) (data as of June 23, 2020)

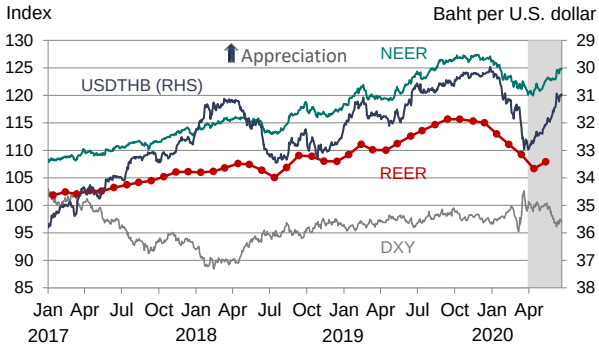
Total corporate financing by instrument*



Note: *Monthly change in outstanding of corporate loans (seasonally adjusted), corporate bonds excluding those issued by commercial banks, and newly issued equities.

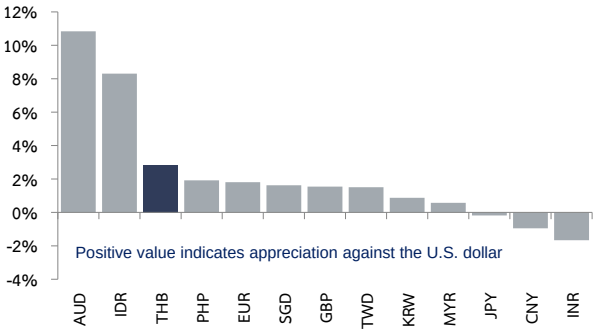
Sources: Bank of Thailand and Thai Bond Market Association (Thai BMA)

The Thai baht vis-a-vis U.S. dollar (USDTHB), Nominal Effective Exchange Rate (NEER), Real Effective Exchange Rate (REER) and the Dollar Index (DXY)



Source: Bank of Thailand and Reuters (data as of June 23, 2020)

Currency movements vis-a-vis the U.S. dollar (Avg. in June 20 compared to avg. in Mar 20)

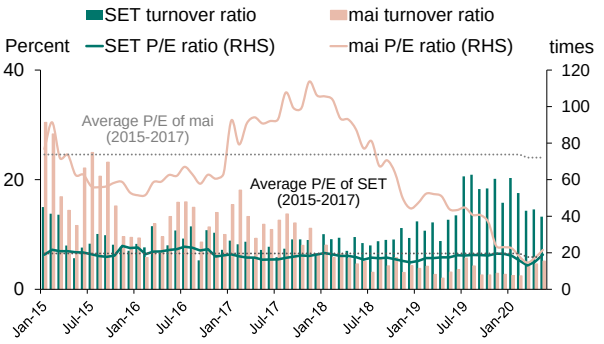


Source: Bank of Thailand and Reuters (data as of June 23, 2020)

Stability: Financial markets

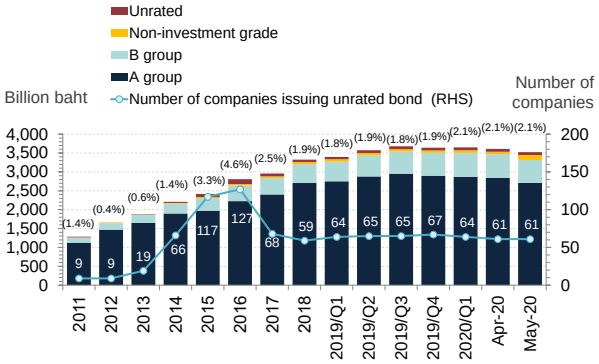
The price-to-earning (P/E) ratio of the Stock Exchange of Thailand (SET) and the Market for Alternative Investment (mai) declined at the end of the first quarter of 2020 due to investors' concerns over the COVID-19 situation and the development of the oil price. However, the ratio returned to the level close to historical average in May. New issuance of corporate bonds softened, as some issuers switched to commercial bank credits following the rise in funding cost in the corporate bond market and the decline in investors' risk appetite. Also, in May 2020, some investment grade issuers in tourism-related sectors affected by COVID-19 received credit rating downgrades, raising the share of non-investment grade bonds.

Current price-to-earning ratio and turnover ratio of SET and mai



Source: Stock Exchange of Thailand (as of May 2020)

Corporate bonds outstanding

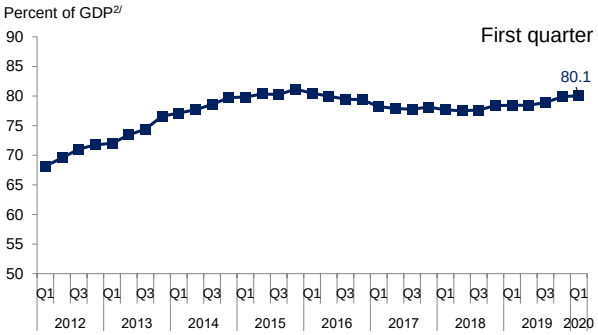


Note: ( ) represents percentage of unrated bonds to total corporate bonds
Source: Thai Bond Market Association (Thai BMA)

Stability: Household sector

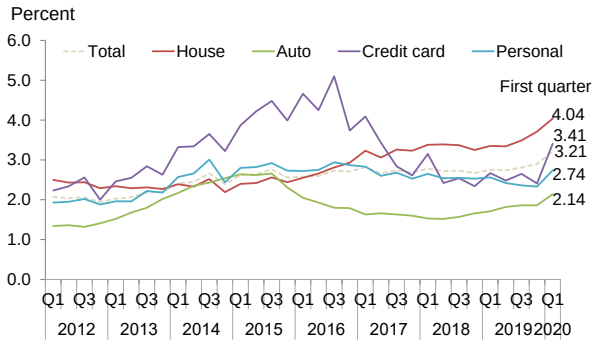
Household financial positions and debt servicing ability deteriorated, as reflected in the household debt-to-GDP ratio in the first quarter of 2020, which rose slightly from the already-high level. Also, NPL ratio (Stage 3) increased in all types of loans, especially credit card loans, partly as a result of COVID-19 impact on economic activities.

Household debt[1/]



Note: [1/] Loans to households by financial institutions
[2/] Calculated by averaging the 4 latest quarterly GDP
[3/] Household debt and GDP data are revised. This results in the different debt to GDP ratios compared to the last MPR.
Source: Bank of Thailand

Share of non-performing loans (NPL) in consumer loans, classified by loan type

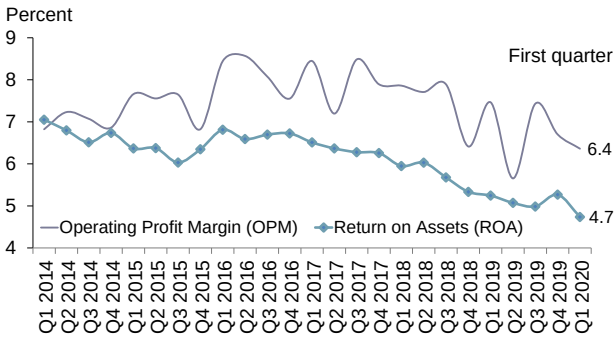


Source: Bank of Thailand

Stability: Corporate Sector

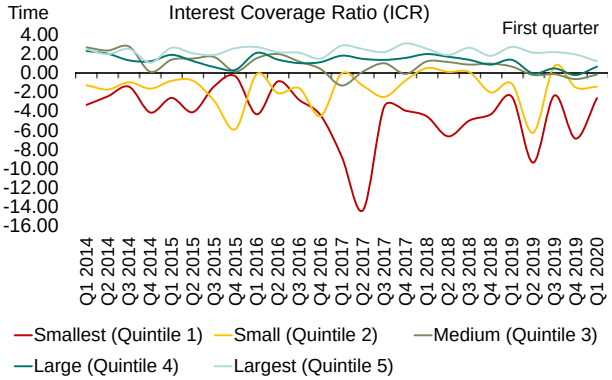
Overall vulnerabilities in the corporate sector increased in the first quarter of 2020, especially among large corporates impacted by oil price decline and merchandise export contraction. This was reflected by deteriorating debt servicing ability of large corporates, including in the return on assets (ROA) and operating profit margin (OPM) of listed companies. The rising NPL among SMEs, particularly in manufacturing, trading, as well as hotel and restaurant sectors, should be closely monitored.

Operating Profit Margin (OPM) and Return on Assets (ROA)*



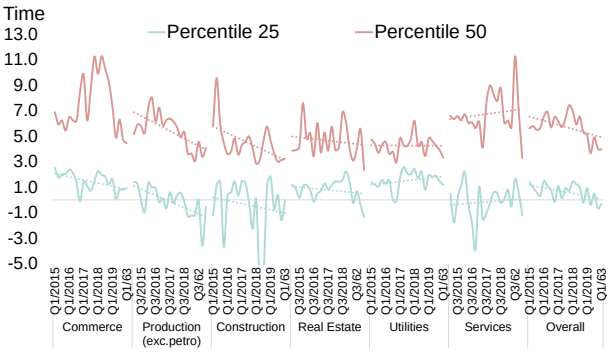
Note: * Median estimates; ROA is returns to average assets.
OPM is operating profits to total sales.
Source: Stock Exchange of Thailand, calculation by Bank of Thailand

Debt serviceability at 25th percentile of each group of firm size



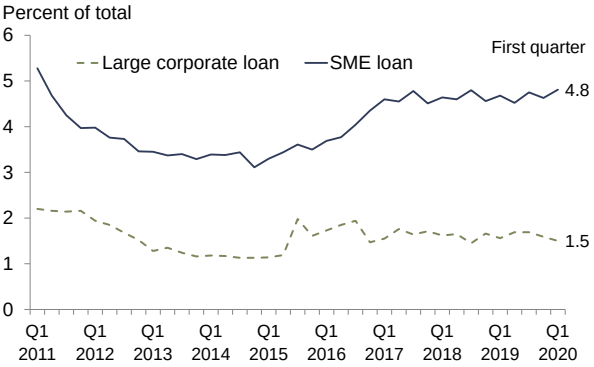
Source: Stock Exchange of Thailand, calculation by Bank of Thailand

Interest Coverage Ratio, classified by sectors



Note: * production exclude Petroleum and chemicals
Source: Stock Exchange of Thailand, calculation by Bank of Thailand

Share of non-performing loans (NPL) in corporate loans, classified by size

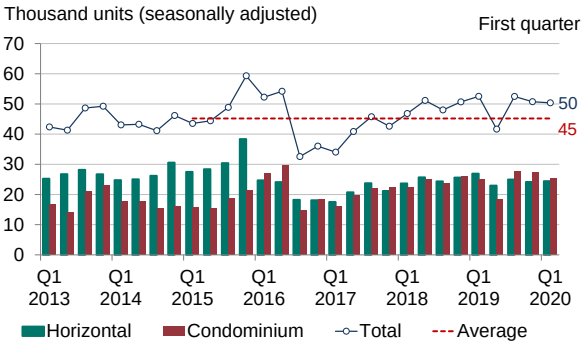


Source: Bank of Thailand

Stability: Real estate sector

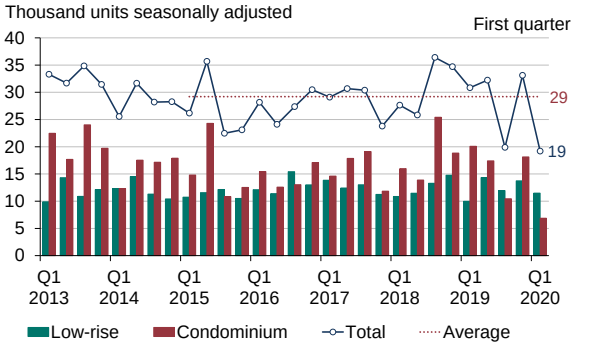
The real estate sector faced a risk from the continuous decline in demand for residential properties from domestic and foreign buyers amidst the COVID-19 pandemic. Developers thus delayed launches of new projects in the first quarter of 2020, especially for condominium projects.

Residential Transfer units in Bangkok and its vicinity



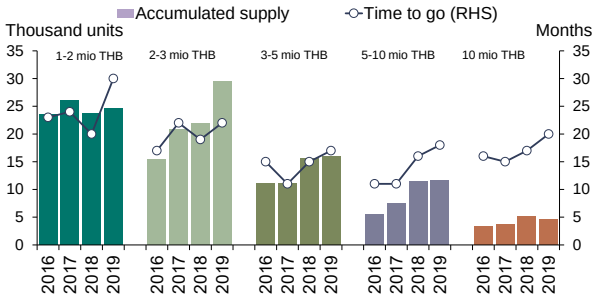
Source: Real Estate Information Center
Note: *Average during 2014-2018

New residential projects launched in Bangkok and its vicinity



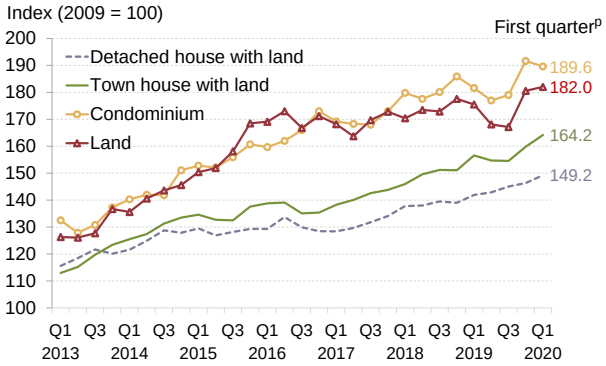
Source: Agency for Real Estate Affairs (AREA), calculation by Bank of Thailand
Note: *Average during 2014-2018

Condominium inventory in Bangkok and vicinity and 'Time to go'



Note: 'Time to go' is the time taken for all real estate inventory to be sold out at the average sales rate per month (since projects launched) given no additional supply.
Source: AREA and calculation by the Bank of Thailand

Real estate price indices

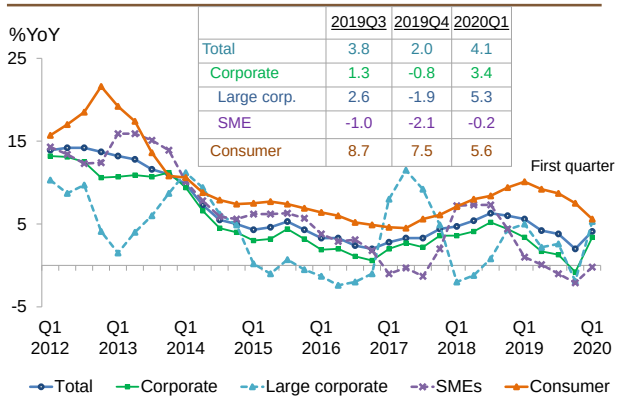


Note: Calculation based on commercial bank loan data
Source: Bank of Thailand

Stability: Financial institutions

The financial position of financial institutions remained robust, as reflected by adequately high levels of commercial banks' reserves and capital buffers to cushion against risks of credit quality deterioration. Nevertheless, overall NPL ratios in the commercial banking system rose in the first quarter of 2020, particularly among consumer loans, in line with economic conditions.

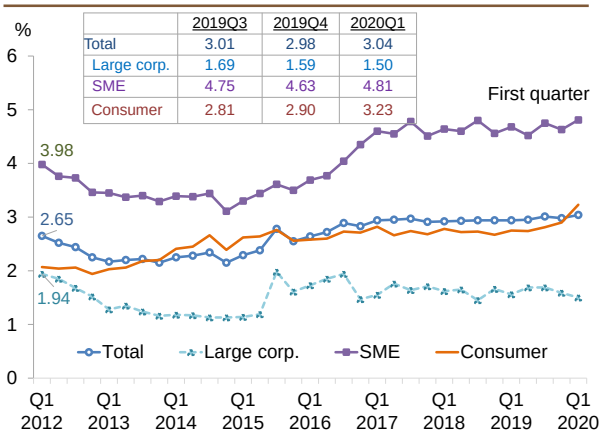
Credit growth in the commercial bank system



	2019Q3	2019Q4	2020Q1
Total	3.8	2.0	4.1
Corporate	1.3	-0.8	3.4
Large corp.	2.6	-1.9	5.3
SME	-1.0	-2.1	-0.2
Consumer	8.7	7.5	5.6

Source: Bank of Thailand

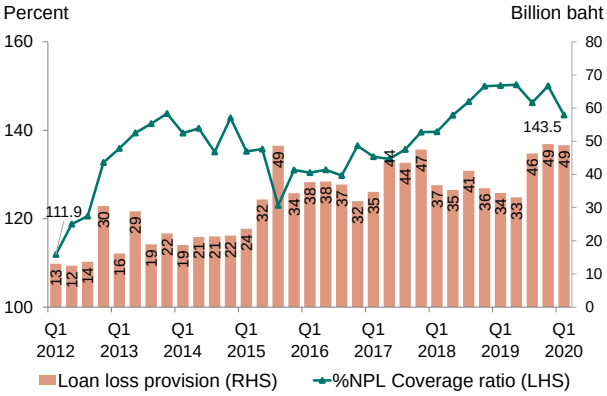
Non-performing loan (NPL)



	2019Q3	2019Q4	2020Q1
Total	3.01	2.98	3.04
Large corp.	1.69	1.59	1.50
SME	4.75	4.63	4.81
Consumer	2.81	2.90	3.23

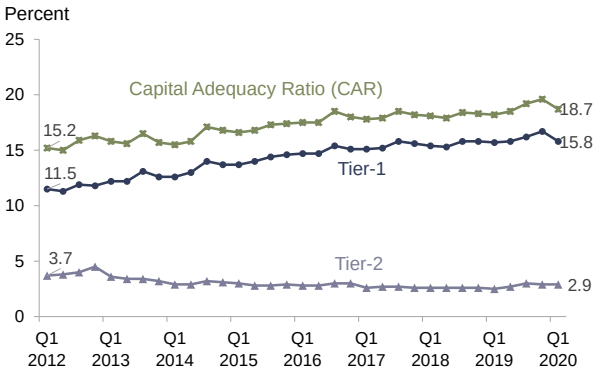
Source: Bank of Thailand

Provisions in commercial bank system



Source: Bank of Thailand

Capital buffers in commercial bank system

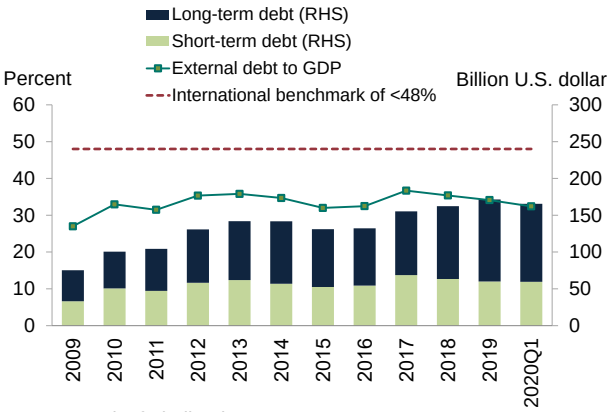


Source: Bank of Thailand

Stability: external positions

Thailand's external stability remained solid, as reflected in the high level of international reserves and the low level of external debt to GDP ratio compared to international benchmark. This would help the Thai economy stay resilient against volatilities in the global financial market.

Thailand's external debt



Source: Bank of Thailand

Reserve to short-term debt

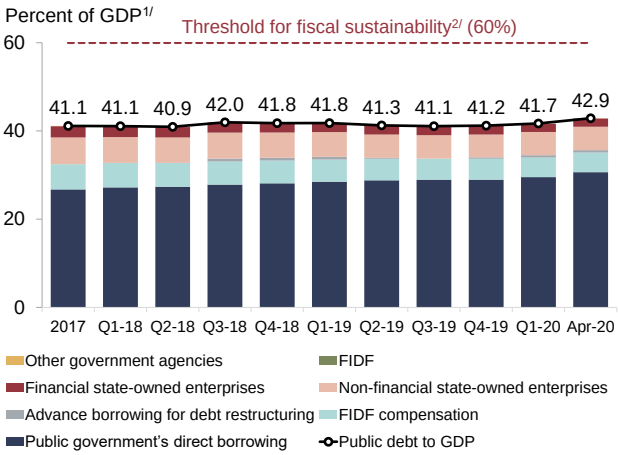


Source: Bank of Thailand

Stability: Fiscal sector

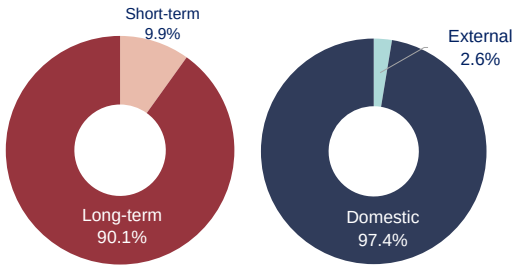
Fiscal stability remained sound. The ratio of public debt to GDP remained below the fiscal sustainability threshold.

Public debt to GDP



Note: 1/ Calculated from GDP by Chain Volume Measure method
2/ Set by Monetary and Fiscal Policy Committee by the authority of State Fiscal and Financial Disciplines Act, B.E. 2561 (2018)
Source: Public Debt Management Office

Outstanding debt as of April 2020



Note: Share of short-term and long-term debt calculated from remaining duration until maturity
Source: Public Debt Management Office



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